THE SPECTRE KNOT: ONE ENDLESS STRAND ON AN APERIODIC MONOTILE

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Abstract. The Spectre of Smith, Myers, Kaplan and Goodman-Strauss tiles the plane, but only non-periodically. We decorate its straight-edged form Tile(1, 1) with fourteen ribbon pieces, the same on every tile, so that ribbons continue straight across every tile edge. We prove that the pieces never close up: in every tiling by copies of Tile(1, 1) of one handedness, including tilings with faults, the ribbons contain no closed strand (the Endless Strand Theorem). In the nest tiling, grown from a single tile by repeating the substitution so that each block is child 1 (a Δ-supertile) of the next, all the pieces of all the tiles join into a single strand, a two-way infinite path that passes through every tile (the Ariadne Theorem). We do not claim this for other tilings, and computations indicate that it fails in some. The proof shows that every supertile, at every level, connects the ribbon ends on its boundary in the same simple pattern: a hook at every boundary vertex but two, and one crossover. When supertiles are glued, only the parities of the shared boundary lengths matter, and these parities are the same at every level, because every side length is a Fibonacci number, twice a Fibonacci number, or a Fibonacci number plus or minus two. This reduces the infinite problem to finitely many cases, which we check by computer, in exact arithmetic. The proof, the programs and this text were produced with an AI system; the author has not verified every step, and the paper has not been refereed (see *Use of AI and verification status* in Section 1).

1. Introduction

A *Celtic knot* is a pattern of interlaced ribbons [2, 5]. How many separate strands a finite design of this kind contains has been studied, for instance, by Fisher and Mellor [7]. A *knot tile* is a polygon carrying ribbon segments that end on its edges, so that copies of the tile, placed edge to edge, link up into longer ribbons. We ask for a knot tile whose ribbons, in a tiling of the whole plane, form one endless strand. In a periodic tiling this cannot happen: if the decorated tiling is invariant under a non-zero translation, its ribbons never form a single strand (Remark 3.5). So a tile that only tiles non-periodically is the natural place to look.

In 2023 Smith, Myers, Kaplan and Goodman-Strauss described the *hat*, an aperiodic monotile [16], and then the *Spectre* [17]: a shape that tiles the plane, admits only non-periodic tilings, and admits only tilings in which all tiles have the same handedness, even when reflections are allowed [17, Thm. 2.2]. For background on tilings and substitution see [9, 1]. The Spectre is a curved-edge form of the polygon Tile(1, 1) (Fig. 1): tilings by Spectres correspond exactly to the tilings by Tile(1, 1) in which all tiles have the same handedness [17, §2].

This paper describes a knot tile on the Spectre and proves that it is *mono-stranded* in the following sense: no tiling contains a closed strand (the Endless Strand Theorem), and in one explicit tiling all its pieces form a single strand (the Ariadne Theorem). We do not claim one strand for every tiling; computations indicate that some tilings have three (Remark 9.1). The decoration (Fig. 1) puts one crossing point on each of the fourteen edges and joins the 28 ribbon ends inside the tile by fourteen pieces: twelve *hooks*, each turning back round

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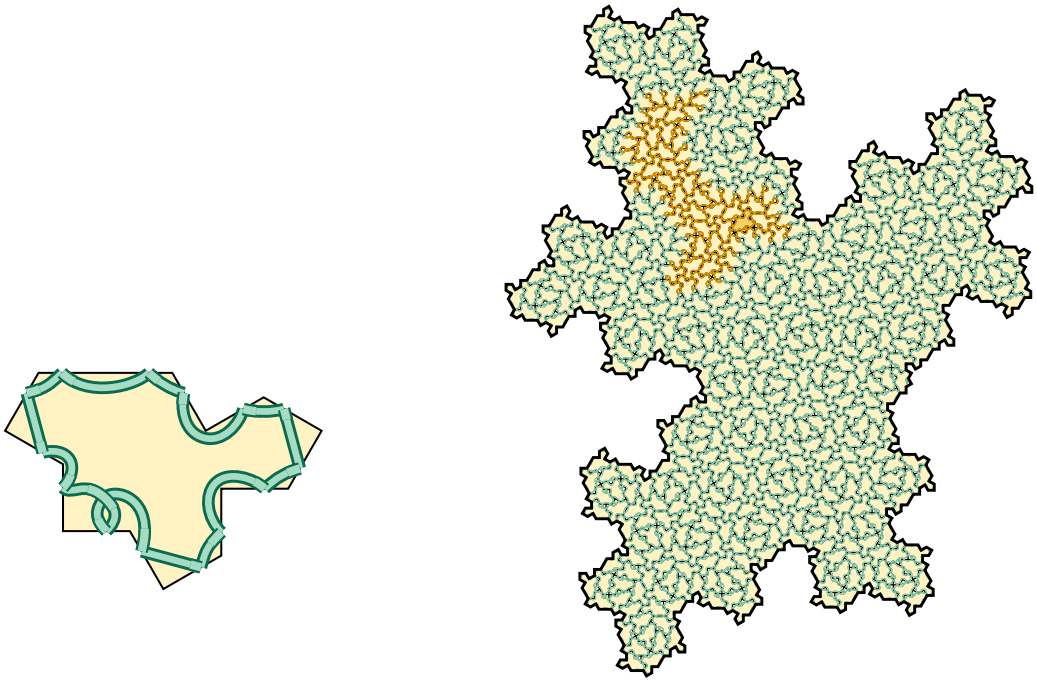


FIGURE 1. Left: the tile Tile(1, 1) with its fourteen ribbon pieces. Right: a supertile of 559 tiles (the block N_3 of Section 8). The original tile N_0 is shaded; all fourteen of its pieces already lie on a single arc (a path of ribbon from rim to rim) of this block, drawn in orange.

one vertex, and two pieces that cross each other inside the tile. Across an edge, the ribbons simply continue straight through the crossing point.

Our first result, the *Endless Strand Theorem*, concerns all tilings. Every tiling by Tile(1, 1) with tiles of one handedness (a *homochiral* tiling; Section 2) groups uniquely into a hierarchy of larger and larger *supertiles* [17, Thm. 2.2]. Either any two tiles lie in a common supertile, or the tiling splits into two or more infinite supertiles, separated by a *fault* (Section 7.3); the Endless Strand Theorem covers both cases.

Endless Strand Theorem. *In every homochiral tiling of the plane by* Tile(1, 1)*, with every tile carrying the decoration of Fig. 1, there is no closed strand: every strand is a two-way infinite path of ribbon pieces.*

Our second result, the *Ariadne Theorem*, gives the single strand; it is named after Ariadne's thread, which led through the whole labyrinth. We prove it for one specific tiling, the *nest tiling* $\mathcal{N}$, which is grown from a single tile N_0 by repeatedly taking the Δ-supertile N_{b+1} that has N_b as its child number 1 (Section 8). In the usual language of substitution tilings, $\mathcal{N}$ is the tiling generated from one tile as a seed by iterating the substitution about it, a fixed point of the substitution (in the combinatorial sense; compare [1]); in terms of the supertile addresses of [18], it is the infinite-level supertile whose sequence of child numbers is $1, 1, 1, \ldots$. Figures 1, 2, 9 and 11 show blocks of it.

Ariadne Theorem. *In the nest tiling $\mathcal{N}$, with every tile carrying the decoration of Fig. 1, all the ribbon pieces of all the tiles form a single strand. It is a two-way infinite path that passes through every tile. More precisely, for every $b \geq 1$, all the arcs of the block N_b (the paths of ribbon through N_b from rim to rim; Section 4) lie on a single arc of N_{b+4}, and of N_{b+3} when b is even.*

Figure 2 shows the merging for N_0: its fourteen pieces lie on 5, 2 and 1 arcs of N_1, N_2 and N_3. Since the curves cross only transversally and in pairs, over and under can be assigned

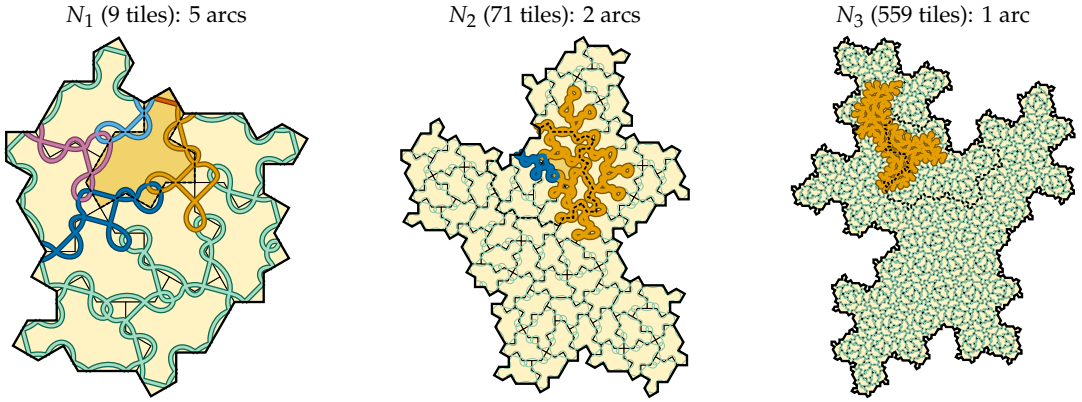


FIGURE 2. The arcs of the nest blocks N_1, N_2, N_3 that contain pieces of the original tile N_0 (shaded), one colour per arc. The dashed outline is the previous block.

alternately, as in Celtic knotwork (Remark 3.6); the single strand then becomes an alternating diagram of a line.

The method. The proof has three ideas.

(1) *Black boxes.* Seen from outside, a supertile is just a list of which ribbon end on its boundary is joined to which. We show that every supertile of every kind and level has the same *signature*: a hook at every boundary vertex but two adjacent ones, A and B, where the two ribbons cross over; and no closed strand inside (Definition 4.3, Theorem 7.3).
(2) *Zigzags and parity.* When two such boxes are glued along a common boundary path, the ribbons zigzag along it, and the outcome depends only on the parity of its length (Lemma 5.1). Gluing therefore reduces to a finite walk on a small combinatorial object, the *skeleton* (Section 5).
(3) *Fixed parities.* The supertiles are combinatorially hexagons and compose by the substitution of [17, Fig. 5.1]. Their side lengths are given by Fibonacci numbers (Proposition 6.2), and their parities are the same at every level. So, for each kind, the skeleton takes only two values, one for odd and one for even levels, and an induction runs for ever.

For the Endless Strand Theorem one more idea is needed: a closed strand that is short compared with the sides of large supertiles, and passes between them, would have to turn back somewhere; only a crossover makes a ribbon turn back (the Hairpin Lemma), and the crossovers of different supertiles are far apart (the Lonely Crossover Lemma; Section 7.3). The merging of all ribbons into one strand (Theorem 8.5) follows the pattern of the induction: we follow the arcs of the blocks by labelling them, and this bookkeeping depends on the level only through finitely many comparisons of side lengths, all of which follow from simple lower bounds on the side lengths.

Computer assistance. The proof rests on finitely many computer checks, all exact (with coordinates in $\mathbb{Z}[e^{i\pi/6}]$, or with integers and rationals): the signatures of the Mystic and of the level-1 supertiles, with their vertices, side lengths and crossover corners; the 18 skeleton walks of the induction step; and a symbolic run of the merging bookkeeping, whose linear conditions on side lengths all follow from lower bounds on the side lengths. The composition pattern is cited from [17] and compared with its figures (on rendered images). Each of these computations was done by two separate programs. Appendix A lists every computation with its script; the programs, with logs of a full run, are published with this paper at `https://cortado-j.github.io/SPECTRE_KNOT/` (folder `code/`).

Scope. The Endless Strand Theorem covers every homochiral tiling, including tilings with faults. The Ariadne Theorem is proved for the nest tiling only, and we make no claim about the number of strands in any other tiling. We do not expect the single strand in every tiling: computations indicate that in at least one other tiling, one without faults, the pieces form exactly three strands (Remark 9.1). See Section 9.

Related work. Curves drawn on tiles and joined across edges appear in several settings; each curve is then closed or two-way infinite, as in Lemma 3.4. For Truchet tilings the closed curves of random tilings have been studied by percolation and renormalisation methods [14, 10]; on rectangular grids the number of components of the mirror curves behind Celtic designs is computable, and without internal mirrors it is the greatest common divisor of the side lengths [11]. On the Penrose rhombs, the arc decoration of Conway and Penrose gives closed curves apart from at most two infinite ones (see [8] and the references there), and Girih interlace on a Penrose tiling produces many closed loops and knots [4]. Single curves through all the tiles of finite patches have been built as Hamiltonian cycles on Ammann–Beenker tilings [15] and as space-filling curves from decorations of substitution tilings [13]. Closest to the present work, [3] describes, in one nested tiling by Tile(1, 1), a single non-crossing curve through every tile, using selected edges. Here every edge carries a ribbon end, the pieces cross, and the single strand is a knot diagram in the sense of Celtic knotwork. We are not aware of earlier full knotwork decorations that form a single strand.

Use of AI and verification status. The author came up with the original idea of looking for a mono-stranded aperiodic knot, and then used an AI system (Claude, by Anthropic) to search for it. The search produced 34 candidate designs; the author chose design 0.1:01 among them and drew its ribbons by hand in a tile editor. At the author's direction the AI system also wrote all the programs, developed the proof and drafted this paper.

The author has not checked every step of the proof, and parts of it are beyond the author's expertise. The paper has not been refereed, and no mathematician has reviewed it, so its main result should be regarded as a claim, not as an established theorem. The AI system reviewed its own argument in several passes, and the errors it found were corrected. All the computations have been re-run in full; the programs and their logs are published with the paper. Where this paper calls a program or a check *independent* or *separate*, it means a separate implementation; all the programs were written by the same AI system.

Outline. Section 2 recalls the facts about Spectre tilings that we use, in the notation of [17]. Section 3 defines the decoration and its strands. Section 4 introduces boxes and the signature. Section 5 proves the Zigzag Lemma and the reduction to skeletons. Section 6 computes the side lengths. Section 7 proves the signature at every level and the Endless Strand Theorem. Section 8 constructs the nest tiling and proves the Ariadne Theorem. Section 9 discusses scope and open questions. Appendix A lists every computation and the program that performs it.

2. Spectre tilings

2.1. **The tile.** Let $\rho = e^{i\pi/6}$. The polygon Tile(1, 1) has fourteen edges of unit length; we label its vertices $v_0, \dots, v_{13}$ counter-clockwise and its edges $e_i = v_i v_{i+1}$ (indices mod 14), with $v_0 = 0$ and

$$v_{i+1} - v_i = \rho^{k_i}, \qquad (k_0, \dots, k_{13}) = (0, 10, 1, 3, 0, 2, 5, 7, 4, 6, 6, 8, 11, 9).$$

All vertices lie in $\mathbb{Z}[\rho]$, and the edge directions are multiples of 30°; see Fig. 3. This is the equilateral member of the family Tile(a, b) [17, §§1–2], with its long side split into the collinear edges e_9, e_{10}, so that v_{10} has interior angle 180°; it is the mirror image of the polygon drawn in [17, Fig. 1.1], and has the handedness of the single Spectre and of the Mystic (a pair of tiles, defined below) drawn in [17, Fig. 2.1]. The clusters of [17, Figs. 2.1, 4.1] are drawn with tiles of the opposite handedness, so our level-1 supertiles (Section 2.2) are their

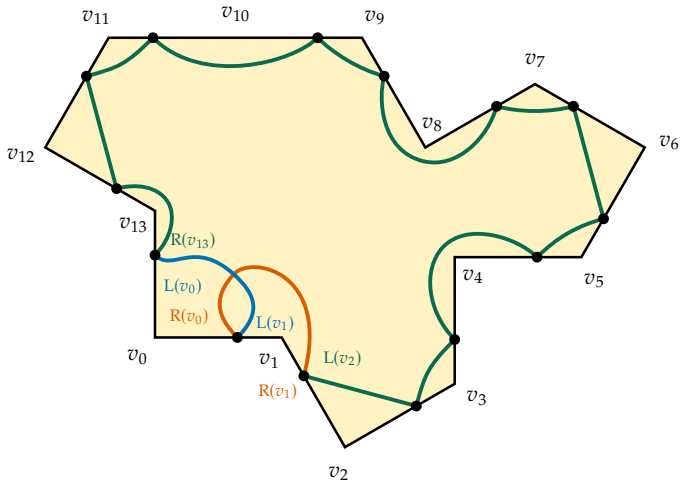


FIGURE 3. The decoration on the true geometry (centre lines of the ribbons of Fig. 1). Each edge carries one crossing point (dot), at which two pieces of the tile end: one leaves towards each end of the edge. On e_i these two ends are the ports $\mathrm{R}(v_i)$ (towards v_i) and $\mathrm{L}(v_{i+1})$ (towards v_{i+1}); the labels show them round the crossover corners. Green: the twelve hooks $\mathrm{L}(v_i) \leftrightarrow \mathrm{R}(v_i)$. Vermilion: $\mathrm{R}(v_0) \leftrightarrow \mathrm{R}(v_1)$; blue: $\mathrm{L}(v_0) \leftrightarrow \mathrm{L}(v_1)$, the crossover at the corners $A = v_0$ and $B = v_1$.

mirror images; the handedness of the level-m supertiles alternates with m [17, Thm. 5.1]. *Spectres*, obtained by replacing every edge of Tile(1, 1) by a suitable curve, have tilings that correspond exactly, with the same placing isometries, to the *homochiral* tilings by Tile(1, 1), in which any two tiles are related by a rotation and a translation [17, §2, Lemma 2.1, p. 5]; so what [17] proves for tilings by Spectres holds for them, and we use it in that form. Reflecting the whole plane if necessary (a reflection maps strands to strands), we assume that every tile is the image of Tile(1, 1) under a rotation and a translation. Such tilings exist [17, Thm. 2.2], and in each the tiles meet full edge to full edge [17, proof of Lemma 2.1]. The tile has no non-trivial symmetry, proper or improper (its cyclic sequence of edge directions has none), so an isometry between two tiles is unique.

2.2. **Supertiles.** We use the hierarchy of [17]. There are nine *kinds*

$$\mathcal{K} = \{\Gamma, \Delta, \Theta, \Lambda, \Xi, \Pi, \Sigma, \Phi, \Psi\}.$$

A *level*-0 *supertile* of kind $K \neq \Gamma$ is a single tile; a level-0 supertile of kind Γ is a *Mystic* [17, Fig. 2.1], the pair of tiles $\{\mathrm{Tile}(1,1),\ g(\mathrm{Tile}(1,1))\}$ with $g(z) = \rho z + v_8$, and its congruent copies; we call Tile(1, 1) its *first* tile. (The Mystic has no non-trivial proper symmetry: one exchanging its tiles would be an involution taking Tile(1, 1) to $g(\mathrm{Tile}(1,1))$, but the only proper motion doing so is g, a rotation by 30°. So the first tile is determined by the pair.) The supertiles of a tiling are those of its hierarchy (Theorem 2.1(a) below): a level-$(m+1)$ supertile is the union of level-m supertiles, its *children*, eight of them, or seven for kind Γ, of the kinds listed in Table 1, arranged as in Proposition 2.4. A level-1 supertile has the kind of its cluster in [17, Fig. 4.1], which is determined by the positions of its six vertices (defined below) [17, proof of Thm. 4.1]; a level-$(m+1)$ supertile, $m \geq 1$, has kind T if its children form the panel T of [17, Fig. 5.1]. For $m \geq 1$, child c is the child at the position given in column c of the last row of Table 1; at level 1 the children are numbered as in Fig. 4. (This is the order used by Kaplan's implementation of the key-point construction of [17, App. A], see [12].) For definiteness a single tile gets the kind of its position in its level-1 supertile;

parent T	0	1	2	3	4	5	6	7
Γ	Π	Δ	Θ	Σ	Ξ	Φ	Γ	
Δ	Ξ	Δ	Ξ	Φ	Σ	Π	Φ	Γ
Θ	Ψ	Δ	Π	Φ	Σ	Π	Φ	Γ
Λ	Ψ	Δ	Ξ	Φ	Σ	Π	Φ	Γ
Ξ	Ψ	Δ	Π	Φ	Σ	Ψ	Φ	Γ
Π	Ψ	Δ	Ξ	Φ	Σ	Ψ	Φ	Γ
Σ	Ξ	Δ	Ξ	Φ	Σ	Π	Λ	Γ
Φ	Ψ	Δ	Ψ	Φ	Σ	Π	Φ	Γ
Ψ	Ψ	Δ	Ψ	Φ	Σ	Ψ	Φ	Γ
						top	top	
position in [17, Fig. 5.1]	left	Δ	bottom	right	Σ	right	left	Γ

TABLE 1. The children of each kind of supertile, and their places in the eight-hexagon layout of [17, Fig. 5.1] (Δ, Σ, Γ: the hexagons of those kinds). For $T = \Gamma$ there is no child at the bottom, and children $2, \dots, 6$ are at the right, the Σ, the top right, the top left and the Γ.

only the distinction between Γ and the other kinds is used at level 0. In particular child 1 of every supertile is a Δ, and the last child is a Γ; at levels $m \geq 1$ a Δ occurs only as child 1.

The nine level-1 supertiles are the Spectre clusters (9 tiles) and the Mystic cluster (8 tiles, kind Γ) of [17, Fig. 2.1], with the markings of [17, Fig. 4.1], reflected (Section 2.1); they are drawn, with tiles of the handedness of Tile(1, 1), in Fig. 4. In the language of [17], our level-1 supertiles are the clusters, which are combinatorially the marked hexagons of [17, Fig. 4.2], and the level-n supertiles of hexagons of [17, §5] are our level-$(n+1)$ supertiles.

The following theorem restates, in our language, results of [17].

Theorem 2.1 (Smith, Myers, Kaplan, Goodman-Strauss [17])**.** *Let $\mathcal{T}$ be a homochiral tiling by* Tile(1, 1).

(a) [17, Thms. 2.2, 3.1, 4.1, §5, Fig. 2.1] *The tiles of $\mathcal{T}$ group uniquely into level-*1 *supertiles. Each contains one Mystic, in the position shown in* [17, Fig. 2.1]*; this Mystic and the other tiles, taken singly, are the level-*0 *supertiles inside it. For every $m \geq 1$ the level-m supertiles of $\mathcal{T}$ group uniquely into level-$(m+1)$ supertiles. Each tile lies in a unique nested sequence $X_0 \subset X_1 \subset \cdots$ with X_m a level-m supertile.*

(b) [17, Thms. 3.1, 4.1, 5.1, §5] *For every $m \geq 1$, the tiling of the plane by the level-m supertiles of $\mathcal{T}$ is combinatorially equivalent (homeomorphic as a topological complex* [17, §2.2]*) to a tiling by regular hexagons, each level-m supertile of kind K corresponding to a hexagon marked as the hexagon K of* [17, Fig. 4.2] *for even m, and as its mirror image for odd m, adjacent supertiles corresponding to hexagons sharing an edge with matching labels; and the grouping into level-$(m+1)$ supertiles corresponds, up to reflection, to the grouping of hexagons shown in* [17, Fig. 5.1].

In (a), [17, Thm. 4.1] gives the grouping into clusters; its uniqueness, like that of the higher levels, is part of [17, Thm. 2.2], and we use it only to speak of *the* hierarchy of $\mathcal{T}$. In (b), the handedness is that of our level-1 supertiles, and it alternates with m (Section 2.1).

In particular every level-m supertile ($m \geq 1$) is a closed topological disc whose boundary, its *rim*, is a simple closed polygon made of tile edges. Its *vertices* are the six points of the rim where it meets two other level-m supertiles, and its *sides* are the six paths along the rim between consecutive vertices. (The tile vertices on a rim are called *rim vertices*; for a supertile of level ≥ 1, "vertex" alone always means one of the six.) A vertex of one supertile that lies on the rim of another is a vertex of that one too, since exactly three supertiles meet there. Each side is shared, as a path of tile edges, with exactly one neighbouring supertile.

child	side 0	side 1	side 2	side 3	side 4	side 5
0	7.1	∂	∂	∂	∂	1.1
1	7.2	0.5	∂	2.0	3.0	4.0
2	1.3	∂	∂	∂	∂	3.1
3	1.4	2.5	∂	∂	∂	4.1
4	1.5	3.5	∂	5.0	6.0	7.3
5	4.3	∂	∂	∂	∂	6.1
6	4.4	5.5	∂	∂	∂	7.4
7	∂	0.0	1.0	4.5	6.5	∂

child	side 0	side 1	side 2	side 3	side 4	side 5
0	6.1	∂	∂	∂	∂	1.1
1	6.2	0.5	∂	∂	2.0	3.0
2	1.4	∂	∂	∂	∂	3.1
3	1.5	2.5	∂	4.0	5.0	6.3
4	3.3	∂	∂	∂	∂	5.1
5	3.4	4.5	∂	∂	∂	6.4
6	∂	0.0	1.0	3.5	5.5	∂

TABLE 2. Adjacency of the children's sides: the entry $c'.j'$ in row c, column j means that side j of child c is side j' of child c'; ∂ means that it lies on the parent's rim. Left: every parent kind except Γ. Right: Γ.

2.3. **Own orientation, side numbering and the crossover corner.** By [17, Thm. 5.1], the supertiles composed from a tiling by marked hexagons are combinatorially equivalent to the *mirror images* of those hexagons. So, at each level, the cyclic order of the labelled sides is reversed, and it is convenient to orient supertiles alternately.

Definition 2.2 (Own orientation)**.** The *own orientation* of a level-m supertile is counter-clockwise if m is even and clockwise if m is odd. We write $\omega_m = (-1)^m$.

Thus a child and its parent are always oriented oppositely. Each supertile has a distinguished rim vertex A, its *crossover corner*, defined recursively: for a tile, $A = v_0$; for a Mystic, A is the vertex v_0 of its first tile Tile$(1,1)$; for a level-$(m+1)$ supertile, A is the crossover corner of its Γ child. (That this point lies on the rim is computed for levels 0 and 1 and follows from Proposition 2.4(d) at higher levels.) The second corner B is the rim vertex next to A in the counter-clockwise direction; it appears in Theorem 7.3 (see Remark 7.4). We call A and B the *crossover corners*. The vertices of a level-m supertile ($m \geq 1$) are numbered $0, \ldots, 5$ in own order, starting with the first vertex at or after A; side j runs from vertex j to vertex $j+1$. Figure 4 shows this numbering for the level-1 supertiles.

2.4. **The composition pattern.** Theorem 2.1(b) says that every level of the hierarchy is assembled in the same combinatorial way. We record this in our numbering.

Definition 2.3 (Composition pattern)**.** The *composition pattern* $\mathcal{C}$ consists of the following data for each parent kind T, given in Tables 1 to 3 and drawn in Fig. 5.

(i) The kinds $K_0, \ldots, K_{n_T-1}$ of the children (Table 1).
(ii) For each child c and each side j of c, either the sibling c' and its side j' that lie across side j, or the symbol ∂ if side j lies on the parent's rim (Table 2).
(iii) For each parent side J, the sequence of child sides (c, j) that make up side J, in the parent's own order (Table 3).

Proposition 2.4 (The pattern holds at every level)**.** *Let $\mathcal{T}$ be a homochiral tiling by* Tile$(1,1)$*, let $m \geq 1$, and let Y be a level-$(m+1)$ supertile of $\mathcal{T}$ of kind T, with children $X_0, \ldots, X_{n_T-1}$. Then:*

(a) *X_c has kind K_c (Table 1);*

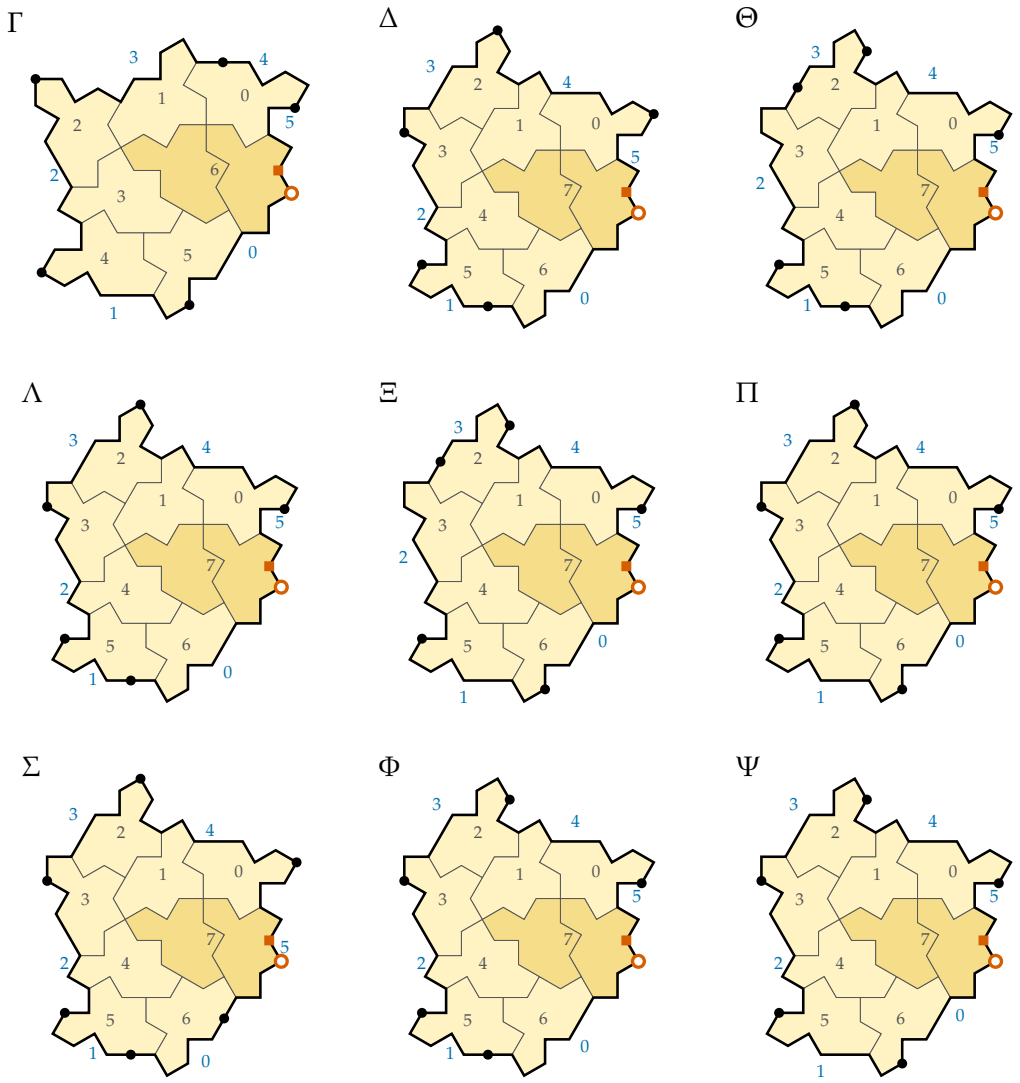


FIGURE 4. The nine level-1 supertiles, with true handedness. The Mystic is the darker pair of tiles. Grey numbers: the children (Table 1); black dots: the six vertices; blue numbers: the sides, numbered clockwise (the own orientation at level 1); open circle: the crossover corner A; filled square: the corner B (Theorem 7.3). For Σ, A lies inside side 5.

(b) *side j of X_c is the same path of tile edges as side j' of $X_{c'}$, traversed in the opposite direction, whenever Table 2 lists $c'.j'$, and it lies on the rim of Y when the table lists ∂;*
(c) *side J of Y is the concatenation of the children's sides listed in Table 3, and vertex J of Y is the vertex $j+1$ of the child c whose side $c.j$ is listed first;*
(d) *the crossover corner A of Y is vertex 0 of its Γ child, which is vertex 0 of Y if $T \neq \Sigma$, and lies inside side 5 of Y if $T = \Sigma$.*

Justification. By Theorem 2.1(b), the level-m supertiles inside Y are combinatorially the hexagons of the corresponding panel of [17, Fig. 5.1]: hexagon edges correspond to sides and hexagon vertices to vertices. We use the following dictionary between the labels of [17, Fig. 4.2] and our side numbers:

(∗) side j of a level-m supertile of kind K carries the label of the edge of hexagon K in direction W, SW, SE, E, NE, NW for $j = 0, \ldots, 5$ respectively (Table 4); its crossover corner A is vertex 0 if $K \neq \Sigma$; if $K = \Sigma$, A is not a vertex but lies in the relative interior of side 5.

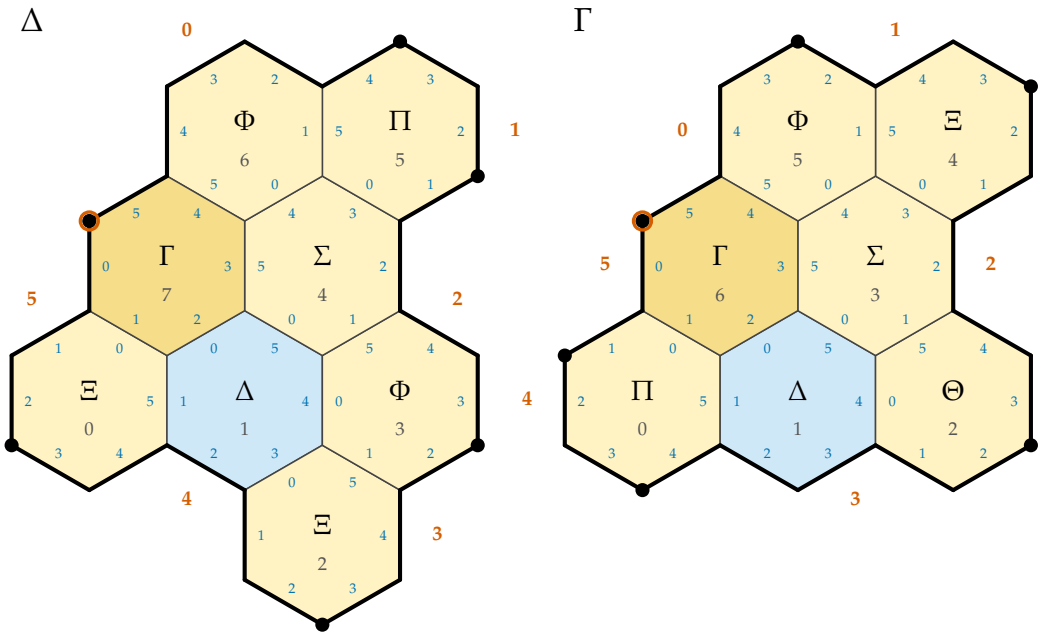


FIGURE 5. The composition pattern C for Δ (left; the same for all kinds except Γ, with the children's kinds as in Table 1) and for Γ (right), in the layout of [17, Fig. 5.1]. Each hexagon is a child: its kind, its number (grey), and its sides 0–5 (blue). Thick black: the parent's rim; black dots: the parent's vertices; vermilion numbers: the parent's sides; circled: the parent's crossover corner, which is vertex 0 of the Γ child. Children's sides are numbered in one rotational sense and the parent's in the other.

We show (∗) by induction on m. For $m = 1$ it is a computation on the level-1 supertiles, which have 44 rim edges for Γ and 46 for the other kinds; their vertices lie at the rim positions, counted from A in own order,

$$\begin{array}{lll} \Gamma : 0, 7, 14, 24, 34, 39 & \Delta : 0, 10, 15, 23, 30, 40 & \Theta : 0, 10, 15, 26, 31, 41 \\ \Lambda : 0, 10, 15, 23, 30, 41 & \Xi : 0, 7, 15, 26, 31, 41 & \Pi : 0, 7, 15, 23, 30, 41 \\ \Sigma : 4, 10, 15, 23, 30, 40 & \Phi : 0, 10, 15, 23, 31, 41 & \Psi : 0, 7, 15, 23, 31, 41. \end{array}$$

Drawn with the edges in odd directions lengthened to $\sqrt{3}$, which turns the tiles into hats and turtles [17, §3], and in the handedness of [17, Fig. 4.1], the nine level-1 supertiles coincide with the nine marked clusters there: the same outlines and the same tiles inside, the vertices at the marked dots, and the labels of Table 4 in the places where that figure has them; A is the dot between the W and NW sides (for Σ, a point of the NW side), on the light tile of the Mystic. A script checks this against a render of the figure (Appendix A). Suppose (∗) holds for the children of Y. The panel draws the inner hexagons with their kinds and edge markings but without their edge labels; the matching rules fix the rotation of each of them, and there is exactly one consistent choice (it is found by the script that encodes the panels, Remark 2.5). It is the same in all nine panels, and Fig. 5 shows the resulting side numbers, so that Tables 2 and 3 can be compared with [17, Fig. 5.1] by eye. With this dictionary, Tables 2 and 3 are then a transcription of the panel of [17, Fig. 5.1] for T, including its vertices (the six marked points on its outline); this gives (a)–(c). The crossover corner of Y is that of its Γ child, which is the child's vertex 0 by (∗), and Table 3 places this vertex at the start of the parent's side 0 if $T \neq \Sigma$ and inside side 5 if $T = \Sigma$; this is (d). Finally, the labels that the panel assigns to the parent's sides, numbered from this point in the parent's own order, are those of Table 4 for every T. So (∗) holds for Y. In particular A always lies on the rim. □

Eight children (every $T \neq \Gamma$): the parent's rim, in own order from vertex 0 of child 7.

position	0	1	2	3	4	5	6	7	8	9	10
segment	7.5	6.4	6.3	6.2	5.4	5.3	5.2	5.1	4.2	3.4	3.3
position	11	12	13	14	15	16	17	18	19	20	21
segment	3.2	2.4	2.3	2.2	2.1	1.2	0.4	0.3	0.2	0.1	7.0

Seven children ($T = \Gamma$), from vertex 0 of child 6; sides $0, \ldots, 5$ begin at positions $0, 3, 6, 11, 16, 18$.

position	0	1	2	3	4	5	6	7	8	9
segment	6.5	5.4	5.3	5.2	4.4	4.3	4.2	4.1	3.2	2.4
position	10	11	12	13	14	15	16	17	18	19
segment	2.3	2.2	2.1	1.3	1.2	0.4	0.3	0.2	0.1	6.0

parent	first position of side					
	0	1	2	3	4	5
Δ	0	5	7	11	14	19
Θ	0	5	7	13	15	20
Λ	0	5	7	11	14	20
Ξ	0	3	7	13	15	20
Π	0	3	7	11	14	20
Σ	2	5	7	11	14	19
Φ	0	5	7	11	15	20
Ψ	0	3	7	11	15	20

TABLE 3. The parent's sides as concatenations of children's sides. A segment $c.j$ is side j of child c, which the parent's own orientation traverses from vertex $j+1$ to vertex j of the child. Side J of the parent consists of the segments from its first position up to the first position of side $J+1$ (cyclically). For $T \neq \Sigma$ the parent's side 0 begins at vertex 0 of the Γ child, which is the parent's crossover corner. For $T = \Sigma$ the crossover corner lies inside side 5 (positions 19–1), after the segments 0.2, 0.1, 7.0.

kind	side 0	side 1	side 2	side 3	side 4	side 5
Γ	α^-	α^+	γ^-	δ^-	β^+	β^-
Δ	γ^+	β^+	ϵ^-	α^+	γ^-	ζ^-
Θ	γ^+	β^+	θ^+	β^+	η	β^-
Λ	γ^+	β^+	ϵ^-	α^+	θ^-	β^-
Ξ	α^-	ϵ^+	θ^+	β^+	η	β^-
Π	α^-	ϵ^+	ϵ^-	α^+	θ^-	β^-
Σ	ζ^+	β^+	ϵ^-	α^+	γ^-	δ^+
Φ	γ^+	β^+	ϵ^-	ϵ^+	η	β^-
Ψ	α^-	ϵ^+	ϵ^-	ϵ^+	η	β^-

TABLE 4. The label of [17, Fig. 4.2] carried by side j of a supertile of each kind: the label of the edge of the hexagon in direction W, SW, SE, E, NE, NW for $j = 0, \ldots, 5$.

Remark 2.5 (Computer checks of the pattern). A script encodes all nine panels of [17, Fig. 5.1], finds the rotations of the inner hexagons used in the proof above, and compares the result with Tables 2 and 3; this is part of the proof. Since the tables carry the whole weight of the citation, we also checked them against supertiles of levels 1 to 8 built with the key-point construction of [17, App. A]. Both agree with the tables (Appendix A).

We shall use one consequence of Theorem 2.1(a) and Proposition 2.4 repeatedly.

Lemma 2.6 (Rigidity)**.** *Any two level-m supertiles of the same kind, in possibly different homochiral tilings, are related by a rotation and translation that maps descendants to descendants with their numbers (children, their children, and so on, down to the tiles). For $m \geq 1$ it maps the six vertices, the sides and the corner A of one onto those of the other.*

Proof. Induction on m. For $m = 0$ this is clear, since a Mystic is a congruent copy of a fixed pair of tiles with its first tile marked. For $m = 1$, the level-1 supertiles of a homochiral tiling are congruent copies of the clusters of Fig. 4 [17, Fig. 2.1, §4], with their vertices at the points marked in [17, Fig. 4.1]; the numbers of their children are those of the reference clusters of Fig. 4, and A is fixed by the Mystic. Let $m \geq 1$. By induction, the children of two level-$(m+1)$ supertiles Y, Y' of kind T are pairwise related by proper motions g_c, which map the vertices of child c of Y onto those of child c of Y' with their numbers. If side j of child c is glued to side j' of child c', then by Proposition 2.4(b), applied to Y and to Y', vertices j and $j+1$ of c coincide with vertices $j'+1$ and j' of c', in Y and in Y'. So g_c and $g_{c'}$ agree on two distinct points; being proper motions, they are equal. The children's adjacency graph is connected, so all the g_c are one motion g. By Proposition 2.4(c),(d), the vertices and the corner A of Y are vertices and the corner A of children, so g maps them, and hence the sides, onto those of Y'. Since g agrees with each g_c, it maps descendants to descendants with their numbers. □

Supertiles built by computer. The computations on tiles (the base case of (∗) and the child numbers at level 1, Proposition 6.1 and Lemma 6.4 at level 1, and Computation 7.2(a)) use the level-1 supertiles built with the key-point construction of [17, App. A]. Their tiles, outlines and vertices are those of the clusters of [17, Figs. 2.1, 4.1] (proof of Proposition 2.4), so, by [17, Thm. 4.1], they are congruent, with their tiles, to the level-1 supertiles of every homochiral tiling. (Our programs find the level-1 vertices as the points where three level-1 supertiles meet inside built supertiles of levels 2 to 4; what certifies them is the comparison with [17, Fig. 4.1].) Higher levels enter the proofs only through the composition pattern.

Write $E_m(K)$ for the number of rim edges (equally, rim vertices) of a level-m supertile of kind K, and $\ell_m(K, j)$ for the number of edges of its side j. By Lemma 2.6 these numbers are well defined.

3. The decoration and its strands

3.1. **Ports and pieces.** Let t be a tile and e one of its edges, with endpoints x and y. The tile has two *ports* on e, the port *near* x and the port *near* y; formally, they are the pairs (e, x) and (e, y). In the drawing (Fig. 3), e carries one crossing point, and the two ribbon ends of t that meet it, one running off towards x and one towards y, are these two ports. For a vertex v_i of the tile, write

$$\mathrm{R}(v_i) = \text{the port on } e_i \text{ near } v_i, \qquad \mathrm{L}(v_i) = \text{the port on } e_{i-1} \text{ near } v_i,$$

the ports just after and just before the corner v_i in counter-clockwise order. Each of the 28 ports is $\mathrm{R}(v_i)$ or $\mathrm{L}(v_i)$ for exactly one i.

Definition 3.1 (The decoration)**.** The *decoration* joins the 28 ports of Tile(1, 1) in pairs by fourteen *pieces*:

$$\underbrace{\mathrm{L}(v_i) \leftrightarrow \mathrm{R}(v_i) \quad (2 \leq i \leq 13)}_{\text{twelve hooks}}, \qquad \underbrace{\mathrm{R}(v_0) \leftrightarrow \mathrm{R}(v_1), \qquad \mathrm{L}(v_0) \leftrightarrow \mathrm{L}(v_1)}_{\text{the crossover}}.$$

Every tile of a homochiral tiling carries the image of this decoration.

In our drawing the two crossover pieces cross exactly once, transversally, and this is the only crossing inside the tile (Remark 3.6). Each hook turns back round one vertex. Figure 3 shows the pieces and their ports, and Fig. 1 the ribbons as drawn. The drawing uses cubic Bézier curves, with the crossing point of edge e_i at distance $0.5 + 0.15(-1)^i$ from v_i. Only the pairing of Definition 3.1 matters for the theorems; any drawing that realises it may be used.

3.2. **The straight-through rule.** Let $\mathcal{T}$ be a homochiral tiling, and let e be an edge shared by tiles t and t', with endpoints x and y.

Definition 3.2 (Straight-through rule)**.** The port of t on e near x is *joined* to the port of t' on e near y, and the port of t near y to the port of t' near x.

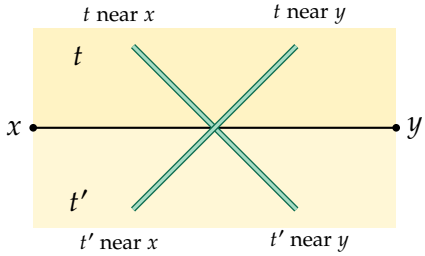


FIGURE 6. The straight-through rule at the crossing point of an edge xy: the end of t that leaves the crossing point towards x continues as the end of t' heading towards y.

This is what a ribbon does when it continues straight through the crossing point (Fig. 6). In the drawing the two tiles' crossing points on e coincide: the tiles traverse e in opposite directions, and an edge e_i of one tile is always glued to an edge e_j of the other with $i + j$ odd. Indeed, by the vertex colouring of [17, proof of Lemma 2.1], the interior angles of Tile$(1, 1)$ at even and odd vertices are multiples of 90° and of 120° respectively, and these classes agree where vertices meet; as v_i of one tile meets v_{j+1} of the other, i and $j + 1$ have the same parity. The proofs use only Definitions 3.1 and 3.2.

3.3. **Strands.**

Definition 3.3 (Strand graph)**.** Let $\mathcal{T}$ be a homochiral tiling. The *strand graph* $\mathcal{G}(\mathcal{T})$ is the multigraph with one vertex for each piece of each tile and one edge for each pair of ports joined by the straight-through rule, between the two pieces having these ports as ends. A *strand* is a connected component of $\mathcal{G}(\mathcal{T})$; a strand is *closed* if it is finite.

Lemma 3.4. *Every vertex of $\mathcal{G}(\mathcal{T})$ has degree* 2*. Hence every strand is either a finite cycle (a closed strand) or a two-way infinite path.*

Proof. A piece has two ends, each a port of its tile on some edge e. In a tiling of the plane, e is shared by exactly one other tile, so each end is joined to exactly one port, an end of a piece of that neighbour. Joins always link pieces of different tiles, so $\mathcal{G}(\mathcal{T})$ has no loops, and every vertex has degree exactly 2. A connected, locally finite multigraph with all degrees 2 is a finite cycle (possibly of length 2) or a two-way infinite path. □

Remark 3.5 (A single strand needs a non-periodic tiling). Let $\mathcal{T}'$ be any edge-to-edge tiling of the plane by compact decorated tiles, each carrying at least one piece, with one crossing point on each edge and strands defined by the straight-through rule, and suppose that $\mathcal{T}'$, decoration included, is invariant under a translation $h \neq 0$. If its strand graph were a single two-way infinite path, with vertices $x \in \mathbb{Z}$, then h would act on it freely, since it moves every tile, and not as a reflection $x \mapsto c - x$, since h^2 fixes no piece; so h would act as $x \mapsto x + k$ with $k \neq 0$, and the pieces would fall into $|k|$ orbits. But the tiles fall into infinitely many orbits, since the cylinder $\mathbb{R}^2/\langle h \rangle$ has infinite area, and every tile has a piece. So a decorated tiling with a translational symmetry never has a single strand. Every tiling by Tile$(1, 1)$ of one handedness is non-periodic [17, Thm. 2.2], so this obstruction does not arise there.

Remark 3.6 (Over and under). In the drawing of Fig. 1 each piece stays inside its tile away from its ends, the two crossover pieces cross once, transversally, no other two pieces of a tile meet except at common ends, and at each crossing point the pieces meet the edge at ±45° (checked numerically; Appendix A). So the curves cross only transversally, in pairs, and the usual chequerboard rule makes over and under alternate along every strand [6]: the single strand of the Ariadne Theorem becomes an alternating diagram of a line, as in Celtic knotwork. None of this affects which pieces lie on which strand.

4. Boxes and the signature

4.1. **Patches and boxes.** A *patch* is a finite set P of tiles of a homochiral tiling whose union is a closed disc with a simple polygonal boundary, the *rim* ∂P. Every supertile of level ≥ 1 is a patch (Theorem 2.1), and so are a tile and a Mystic (whose two tiles share four consecutive edges, $e_6, \dots, e_9$ of the first tile; its rim has 20 edges). Every rim edge belongs to exactly one tile of P. Let $E = E(P)$ be the number of rim edges, and number the rim vertices $u_0, \dots, u_{E-1}$ in a chosen orientation of the rim, indices mod E. For a supertile we use its own orientation (Definition 2.2), except in Section 5, where a parent takes its children's orientation. For a rim vertex u_k define the *rim ports*

$$\mathrm{R}(u_k) = \text{the port on } u_k u_{k+1} \text{ near } u_k, \qquad \mathrm{L}(u_k) = \text{the port on } u_{k-1} u_k \text{ near } u_k,$$

where the port belongs to the tile of P containing that edge. There are $2E$ rim ports. For the tile itself, with its counter-clockwise orientation, these are the ports $\mathrm{R}(v_i)$, $\mathrm{L}(v_i)$ of Section 3. Reversing the orientation exchanges R and L at every vertex.

Lemma 4.1 (The box of a patch)**.** *Start at a rim port of P and follow the pieces of the tiles of P, crossing every interior edge of P by the straight-through rule. The path ends at a different rim port. This defines an involution π_P on the rim ports without fixed points. The pieces of P that are not reached from any rim port form finitely many cycles, the* closed strands of P*; write $\kappa(P)$ for their number.*

Proof. The pieces of P and the joins across interior edges form a finite graph in which every piece has degree 2, except that a piece end at a rim port has no partner. Its components are therefore paths, whose two ends are rim ports, and cycles. Paths have two distinct ends. □

We call the paths of Lemma 4.1 the *arcs* of P; there are exactly E of them. The triple $(\partial P, \pi_P, \kappa(P))$ is the *box* of P. Everything we need about the inside of P is in its box, as the next lemma shows.

Lemma 4.2 (Gluing)**.** *Let a patch P be the union of patches $P_1, \dots, P_n$ with pairwise disjoint interiors. Then the box of P is determined by the boxes of the P_i together with the identification of their shared rim edges. More precisely, π_P is computed by alternately applying the π_{P_i} and the straight-through rule across shared edges, until a rim port of P is reached. The closed strands of P are those of the P_i together with the cycles of this alternation that contain no rim port of P.*

Proof. Every arc or closed strand of P is a concatenation of arcs and closed strands of the P_i, joined across edges shared by two of the P_i. □

4.2. **The signature.**

Definition 4.3 (Signature)**.** Let A and B be adjacent rim vertices of a patch P. We say that P has the *signature at* (A, B) if

(i) $\pi_P(\mathrm{L}(u)) = \mathrm{R}(u)$ for every rim vertex $u \notin \{A, B\}$ (a *hook* at u);
(ii) $\pi_P(\mathrm{R}(A)) = \mathrm{R}(B)$ and $\pi_P(\mathrm{L}(A)) = \mathrm{L}(B)$ (the *crossover*);
(iii) $\kappa(P) = 0$.

The definition is symmetric in A and B and does not depend on the orientation of the rim, since reversing it exchanges L and R. A patch with the signature has $E - 2$ hook arcs, one at each rim vertex other than A and B, and two crossover arcs. Figure 7 is a picture.

Lemma 4.4. *A tile has the signature at (v_0, v_1), and a Mystic has the signature at the vertices v_0, v_1 of its first tile.*

Proof. For the tile this is Definition 3.1. For the Mystic it is a finite computation with Lemma 4.2: there is no closed strand, the 18 rim vertices other than these two are hooks, and the crossover persists (Appendix A). □

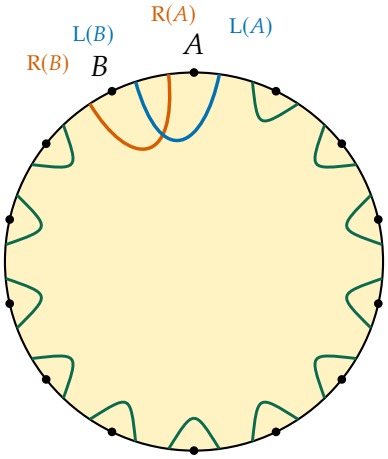


FIGURE 7. The signature at (A, B), drawn for a rim with 14 vertices oriented counter-clockwise, as for the tile ($A = v_0$, $B = v_1$): a hook at every rim vertex except A and B, and the two crossover arcs $\mathrm{R}(A) \leftrightarrow \mathrm{R}(B)$ and $\mathrm{L}(A) \leftrightarrow \mathrm{L}(B)$.

5. The Zigzag Lemma and the skeleton

Throughout this section a patch P is the union of patches (*children*) $X_0, \dots, X_{n-1}$ with disjoint interiors, all of them oriented in the same rotational sense, and P is oriented in that sense too. Each child X_c has the signature at (A_c, B_c). When P is a supertile and the X_c are its children, the children carry their own orientation, so P is oriented opposite to its own orientation. This does not matter for the signature, which does not depend on the orientation (Section 4); but it does exchange L and R at the parent's rim vertices.

5.1. **Runs and the Zigzag Lemma.** Two children $c \neq c'$ oriented in the same sense traverse a common rim path in opposite directions. For a rim vertex u of X_c we write $\mathrm{R}_c(u), \mathrm{L}_c(u)$ for the rim ports of X_c in the common orientation fixed above (for supertiles, their own orientation).

Lemma 5.1 (Zigzag Lemma)**.** *Let $u_0u_1 \cdots u_n$ ($n \geq 1$) be a path of n edges lying on the rims of both X_c and $X_{c'}$, traversed by X_c from u_0 to u_n. Suppose that none of the interior vertices $u_1, \dots, u_{n-1}$ is a crossover corner of X_c or of $X_{c'}$. Then, alternately applying the straight-through rule across the edges u_ku_{k+1} and the hooks of X_c and $X_{c'}$ at $u_1, \dots, u_{n-1}$, the ports at the two ends of the path are joined as follows:*

$$\begin{aligned} &n \text{ even:} && \mathrm{R}_c(u_0) \leftrightarrow \mathrm{L}_c(u_n), && \mathrm{R}_{c'}(u_n) \leftrightarrow \mathrm{L}_{c'}(u_0); \\ &n \text{ odd:} && \mathrm{R}_c(u_0) \leftrightarrow \mathrm{R}_{c'}(u_n), && \mathrm{L}_c(u_n) \leftrightarrow \mathrm{L}_{c'}(u_0). \end{aligned}$$

These two connections use every port on the edges of the path, so no closed strand lies along the path. Moreover, for $0 < k < n$, the hook of X_c at u_k lies on the connection through $\mathrm{R}_c(u_0)$ if k is even. If k is odd, it lies on the connection through $\mathrm{L}_c(u_n)$ when n is odd, and on the connection through $\mathrm{R}_{c'}(u_n)$ when n is even.

Proof. See Fig. 8. Start at $\mathrm{R}_c(u_0)$, the port of X_c on u_0u_1 near u_0. The straight-through rule takes it to the port of $X_{c'}$ on u_0u_1 near u_1. In the orientation of $X_{c'}$, which runs from u_1 to u_0 along this edge, that port is $\mathrm{R}_{c'}(u_1)$. If $n = 1$ we have reached $\mathrm{R}_{c'}(u_n)$. Otherwise u_1 is a hook of $X_{c'}$, which leads to $\mathrm{L}_{c'}(u_1)$, the port of $X_{c'}$ on u_1u_2 near u_1. Crossing u_1u_2 leads to the port of X_c on u_1u_2 near u_2, which is $\mathrm{L}_c(u_2)$. If $n = 2$ we stop. Otherwise the hook of X_c at u_2 leads to $\mathrm{R}_c(u_2)$, and the pattern repeats with all indices shifted by two. So the path visits the hooks of $X_{c'}$ at odd k and the hooks of X_c at even k. It ends at $\mathrm{L}_c(u_n)$ if n is even and at $\mathrm{R}_{c'}(u_n)$ if n is odd. The same argument from the other end, starting at $\mathrm{R}_{c'}(u_n)$ for even n and at $\mathrm{L}_c(u_n)$ for odd n, gives the second connection; it visits the remaining hooks. Each

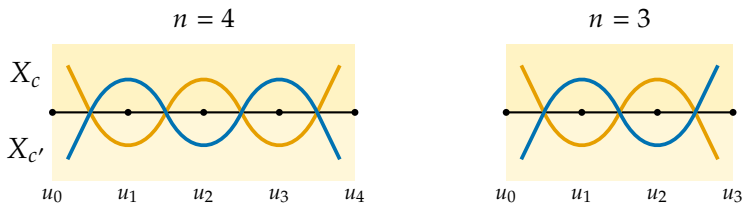


FIGURE 8. The Zigzag Lemma. Along a shared path whose interior corners are hooks on both sides, the ribbons cross the path at every edge and turn back at every corner, alternately in X_c (above) and $X_{c'}$ (below). For even n each connection returns to the side it started from; for odd n it ends on the other side.

connection crosses each of the n edges once, using two of its four ports, so together they use all of them. □

5.2. **Marks, runs and the skeleton.**

Definition 5.2 (Marks and runs)**.** A *mark* is a point that is a rim vertex of some child and is either

(i) a *junction*: a rim vertex of a child X_c at which the two rim edges of X_c have different neighbours, where a neighbour is a sibling or "outside P"; or
(ii) one of the crossover corners A_c, B_c of some child.

A *run* of X_c is a path along the rim of X_c between two consecutive marks lying on that rim. (Each child has at least two marks on its rim, A_c and B_c, so runs are paths with distinct ends.)

Along a run the neighbour of X_c does not change. So a run either lies on the rim of P (a *rim run*) or is shared with exactly one sibling $X_{c'}$ (an *interior run*). In the second case the same path is also a run of $X_{c'}$, traversed in the opposite direction, because marks are points and the same marks bound it on both sides; we call it the *partner* of the run. At an interior vertex u of an interior run, X_c and $X_{c'}$ share both edges at u, so the full angle at u is filled by these two children and no third child touches u. Since crossover corners are marks, the Zigzag Lemma applies to every interior run.

Definition 5.3 (Skeleton)**.** The *skeleton* $\mathcal{S}$ of the composition $P = \bigcup X_c$ consists of the following data: the children and their kinds; for each child, the cyclic sequence of its marks and runs, and which of its marks are A_c and B_c; for each interior run, its partner run and the parity of its length; the cyclic order of the rim runs round ∂P.

Two skeletons of compositions with the same children (by number and kind) are *equal* if there is a bijection of their marks that preserves, for each child, the cyclic order of its marks and the labels A_c, B_c, and preserves partner runs, the parities of interior runs and the cyclic order of the rim runs. The maps τ and σ and the parent ports below are defined from these data alone, so such a bijection carries them, and hence the conclusions of Proposition 5.4, from one composition to the other.

The *reduced ports* are the ports $\mathrm{R}_c(u)$ and $\mathrm{L}_c(u)$ for every child c and every mark u on its rim. The port $\mathrm{R}_c(u)$ lies on the run of X_c that starts at u, and $\mathrm{L}_c(u)$ on the run that ends at u. Define two involutions on reduced ports.

- *Inside a child*: $\tau(\mathrm{L}_c(u)) = \mathrm{R}_c(u)$ if $u \notin \{A_c, B_c\}$; $\tau(\mathrm{R}_c(A_c)) = \mathrm{R}_c(B_c)$ and $\tau(\mathrm{L}_c(A_c)) = \mathrm{L}_c(B_c)$.
- *Across an interior run* from u to u' of X_c, of length n, with partner in $X_{c'}$: if n is even, $\sigma(\mathrm{R}_c(u)) = \mathrm{L}_c(u')$ and $\sigma(\mathrm{R}_{c'}(u')) = \mathrm{L}_{c'}(u)$; if n is odd, $\sigma(\mathrm{R}_c(u)) = \mathrm{R}_{c'}(u')$ and $\sigma(\mathrm{L}_c(u')) = \mathrm{L}_{c'}(u)$.

Both τ and σ are determined by $\mathcal{S}$. The *parent ports* are the reduced ports on rim runs. They are exactly the rim ports $\mathrm{R}(u), \mathrm{L}(u)$ of P at the rim vertices u of P that are marks (the *parent marks*), since X_c and P induce the same orientation on a rim run.

Proposition 5.4 (Reduced composition)**.** *Let $P = \bigcup X_c$, with every child having the signature. Then:*

(a) *every rim vertex of P that is not a mark is a hook of P;*
(b) *for a parent port p, $\pi_P(p)$ is the first parent port among $\tau(p)$, $\tau\sigma\tau(p)$, $\tau\sigma\tau\sigma\tau(p), \ldots$ (the ports $\sigma\tau(p), \ldots$ in between lie on interior runs);*
(c) *$\kappa(P)$ is the number of cycles of the alternation of σ and τ that contain no parent port.*

In particular, π_P at the parent marks and $\kappa(P)$ are determined by the skeleton.

Proof. (a) Let u be a rim vertex of P that is not a mark. The rim edge of P after u lies in some X_c; if the other rim edge of X_c at u were shared with a sibling, u would be a junction. So both rim edges of P at u belong to X_c, and u lies inside a rim run of X_c and is not a crossover corner of X_c. Hence $\pi_{X_c}(\mathrm{L}_c(u)) = \mathrm{R}_c(u)$, and both ports are rim ports of P. (b) and (c) follow from Lemma 4.2: the ribbon starting at p runs through X_c to a reduced port $\tau(p)$, since by the signature of X_c the arc of X_c starting at a port at a mark ends at a port at a mark. If $\tau(p)$ lies on an interior run, the ribbon then follows that run, by the Zigzag Lemma, to $\sigma\tau(p)$. A closed strand of P not inside a child must cross some shared edge, and by the Zigzag Lemma it then passes through a reduced port; following it from there gives a cycle of the alternation. Conversely each such cycle, read as arcs of children joined by zigzag connections, is a closed strand of P, and distinct cycles give distinct strands. The children have no closed strands. □

Corollary 5.5 (Where a hook goes)**.** *In the situation of Proposition 5.4, let u be a rim vertex of X_c that is not a mark. If u lies inside a rim run, the hook of X_c at u is a hook of P. If u lies inside an interior run, traversed by X_c from u_0 to u_n, at distance k from u_0, then the hook of X_c at u lies on the same arc or closed strand of P as the reduced port $\mathrm{R}_c(u_0)$ if k is even; if k is odd, as $\mathrm{L}_c(u_n)$ when n is odd and as $\mathrm{R}_{c'}(u_n)$ when n is even.*

Proof. This is the last part of Lemma 5.1, together with Proposition 5.4(a). □

6. Side lengths

6.1. **Letters.** Every side of a level-m supertile ($m \geq 1$) carries one of the edge labels of [17, Fig. 4.2] (Table 4). Ignoring the signs, there are eight *letters*, $\alpha, \beta, \gamma, \delta, \epsilon, \zeta, \eta, \theta$. Two sides that are glued to each other carry labels x^+ and x^- (or η and η), and so the same letter; this is the matching rule of [17, §4] (Theorem 2.1(b)). For siblings it can also be read off Tables 1, 2 and 4; for supertiles with different parents, in particular across a fault (Section 7.3), we use it as cited.

Proposition 6.1 (Lengths by letter)**.** *For every $m \geq 1$ the length $\ell_m(K, j)$ depends only on the letter of side j of K. Writing x_m for the common length of the sides with letter x,*

$$
\begin{aligned}
\alpha_{m+1} &= \beta_m + \epsilon_m + \eta_m, & \beta_{m+1} &= \alpha_m + \epsilon_m,\\
\gamma_{m+1} &= \beta_m + 2\epsilon_m + \eta_m + \theta_m, & \delta_{m+1} &= \alpha_m + \beta_m + \epsilon_m + 2\theta_m,\\
\epsilon_{m+1} &= 3\epsilon_m + \eta_m, & \zeta_{m+1} &= \alpha_m + \epsilon_m + \theta_m,\\
\eta_{m+1} &= 4\epsilon_m + \eta_m, & \theta_{m+1} &= 4\epsilon_m + \eta_m + \theta_m,
\end{aligned}
\tag{1}
$$

with $(\alpha_1, \beta_1, \gamma_1, \delta_1, \epsilon_1, \zeta_1, \eta_1, \theta_1) = (7, 5, 10, 10, 8, 6, 10, 11)$.

Proof. For $m = 1$ this is a direct count on the level-1 supertiles (Fig. 4; Appendix A). By Proposition 2.4(c), $\ell_{m+1}(T, J)$ is the sum of the lengths of the children's sides listed in Table 3. Inspection of Tables 1, 3 and 4 shows that the multiset of letters of these children's sides depends only on the letter of side J of T, and is the one in (1). For example, side 0 of Δ (letter

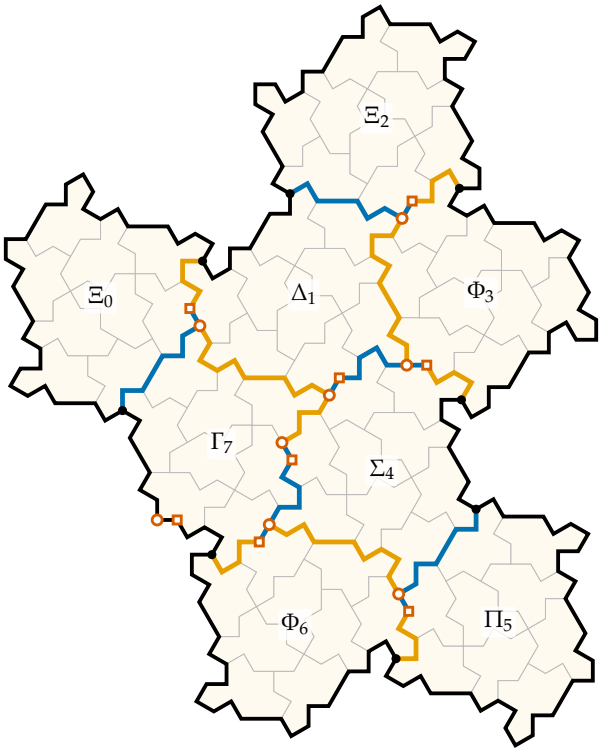


FIGURE 9. Marks and runs for the composition of a level-2 Δ-supertile from its eight level-1 children (subscripts are child numbers). Dots: the 24 marks (open circles and squares: the children's crossover corners A_c and B_c). Shared runs are coloured by the parity of their length: blue odd, orange even. Black: rim runs. The skeleton records this picture up to the lengths of the runs, keeping only the parities of the shared runs.

γ) is made of 7.5, 6.4, 6.3, 6.2, 5.4, with letters $\beta, \eta, \epsilon, \epsilon, \theta$. The claim follows by induction on m. □

The recurrence (1) has a surprisingly simple solution. Let F_n be the Fibonacci numbers, $F_1 = F_2 = 1$.

Proposition 6.2 (Fibonacci side lengths)**.** *For every $m \geq 1$,*

$$\alpha_m = F_{3m+2} + 2, \qquad \beta_m = F_{3m+1} + 2, \qquad \gamma_m = \delta_m = \eta_m = 2F_{3m+2},$$
$$\epsilon_m = F_{3m+3}, \qquad \zeta_m = 2F_{3m+1}, \qquad \theta_m = F_{3m+4} - 2.$$

Proof. For $m = 1$, $(F_4, F_5, F_6, F_7) = (3, 5, 8, 13)$ gives the initial values. For the induction step put $a = F_{3m+1}$ and $b = F_{3m+2}$, so that $F_{3m+3} = a + b$, $F_{3m+4} = a + 2b$, $F_{3m+5} = 2a + 3b$, $F_{3m+6} = 3a + 5b$ and $F_{3m+7} = 5a + 8b$. Substituting into (1):

$$\alpha_{m+1} = (a+2) + (a+b) + 2b = F_{3m+5} + 2, \qquad \beta_{m+1} = (b+2) + (a+b) = F_{3m+4} + 2,$$
$$\gamma_{m+1} = (a+2) + 2(a+b) + 2b + (a+2b-2) = 4a + 6b = 2F_{3m+5},$$
$$\delta_{m+1} = (b+2) + (a+2) + (a+b) + 2(a+2b-2) = 4a + 6b,$$
$$\eta_{m+1} = 4(a+b) + 2b = 4a + 6b, \qquad \epsilon_{m+1} = 3(a+b) + 2b = F_{3m+6},$$
$$\zeta_{m+1} = (b+2) + (a+b) + (a+2b-2) = 2F_{3m+4},$$
$$\theta_{m+1} = 4(a+b) + 2b + (a+2b-2) = 5a + 8b - 2 = F_{3m+7} - 2.$$ □

Since F_n is even exactly when $3 \mid n$:

Corollary 6.3 (Parities are constant)**.** *For every $m \geq 1$ the lengths α_m, β_m, θ_m are odd and γ_m, δ_m, ϵ_m, ζ_m, η_m are even.*

For example, the rim of a level-m Δ-supertile has length $E_m(\Delta) = \alpha_m + \beta_m + 2\gamma_m + \epsilon_m + \zeta_m$, which is 46, 182, 758, 3,198, 13,534, ...

6.2. **The crossover corner.**

Lemma 6.4 (Position of A)**.** *Let $m \geq 1$. For $K \neq \Sigma$ the crossover corner A of a level-m K-supertile is a vertex, hence its vertex 0. For $K = \Sigma$ it lies on side 5 (letter δ), at distance*

$$d_m = 2F_{3m+1}$$

from vertex 5 and at distance $\delta_m - d_m = 2F_{3m}$ from vertex 0.

Proof. For $K \neq \Sigma$ this is Proposition 2.4(d) for $m \geq 2$, and is read off Fig. 4 for $m = 1$. For $m = 1$, $d_1 = 6$ is read off the level-1 Σ-supertile. By Proposition 2.4(d) and Table 3, the crossover corner of a level-$(m+1)$ Σ-supertile follows the children's sides 0.2, 0.1, 7.0 on its side 5. Child 0 is a Ξ and child 7 a Γ, so these sides have letters θ, ϵ, α and $d_{m+1} = \theta_m + \epsilon_m + \alpha_m = (a+2b-2) + (a+b) + (b+2) = 2F_{3m+4}$, in the notation of the previous proof. □

Lemma 6.5 (Lower bounds)**.** *For every $m \geq 1$: the shortest side length at level m is $\beta_m \geq 5$; $d_m \geq 6$; and $\delta_m - d_m \geq 4$. Moreover $d_m \geq \beta_m$, and for $m \geq 2$ also $\delta_m - d_m \geq \beta_m$.*

Proof. By Proposition 6.2, $\alpha_m - \beta_m = F_{3m}$, $\zeta_m - \beta_m = F_{3m+1} - 2$ and $\epsilon_m - \beta_m = F_{3m+2} - 2$ are positive, and $\gamma_m = \delta_m = \eta_m = 2F_{3m+2}$ and $\theta_m = F_{3m+4} - 2$ exceed $\beta_m = F_{3m+1} + 2$; so β_m is the shortest, and $\beta_m \geq \beta_1 = 5$. Also $d_m \geq d_1 = 6$ and $\delta_m - d_m = 2F_{3m} \geq 2F_3 = 4$. Finally $d_m - \beta_m = F_{3m+1} - 2 > 0$, and for $m \geq 2$ $\delta_m - d_m - \beta_m = 2F_{3m} - F_{3m+1} - 2 = F_{3m-2} - 2 > 0$. □

7. The signature at every level, and no closed strands

7.1. **The skeleton as a function of side lengths.** Fix a parent kind T. Let λ assign a positive integer $\lambda(x)$ to each letter x, let d be an integer with $0 < d < \lambda(\delta)$, and let $\omega \in \{\pm 1\}$. Build an abstract composition from Tables 1 and 2. The children are $c = 0, \ldots, n_T - 1$; the rim of child c is a cycle of six sides, side j having length λ(letter of side j of K_c). The child's crossover corner A_c is its vertex 0, or for $K_c = \Sigma$ the point of its side 5 at distance d from vertex 5, and $B_c = A_c + \omega$, the rim vertex one step from A_c in the child's own order, forwards if $\omega = 1$ and backwards if $\omega = -1$. Sides are identified in pairs as in Table 2, reversing direction: vertex j of c with vertex $j' + 1$ of c', and the point at distance o from vertex j with the point at distance $\lambda - o$ from vertex j'; paired sides have the same letter and hence the same length λ. The sides marked ∂ form the rim, in the order of Table 3. Marks, runs and the skeleton are then defined as in Section 5: junctions are the children's vertices at which the neighbour in Table 2 changes. Call the resulting skeleton $\mathcal{S}_T(\lambda, d, \omega)$.

The abstract composition describes the actual ones, in the following sense.

(**) If the children of a level-$(m+1)$ T-supertile ($m \geq 1$) have the signature at $(A_c, A_c + \omega_m)$ in their own orientation, then its composition (with all children in their own orientation) is the abstract composition for (ℓ_m, d_m, ω_m), where ℓ_m also denotes the vector of letter lengths: the children correspond with their numbers and kinds, the point at distance o from vertex j along side j of a child corresponding to the point so described, and glued points, the corners A_c, B_c, the junctions and the parent's rim with the positions of Table 3 correspond. In particular the skeleton of its composition is $\mathcal{S}_T(\ell_m, d_m, \omega_m)$.

This follows from Propositions 2.4 and 6.1 and Lemma 6.4: (i) by Proposition 2.4(a),(b) and Proposition 6.1 the children have the kinds and side lengths of the abstract composition, and side j of X_c is side j' of $X_{c'}$ traversed oppositely, so points are matched as above; (ii) the rims of two children meet only in the closed sides they share: at a point inside a side both rim edges are shared with the sibling across that side, so the two children fill a neighbourhood,

and a vertex lies on exactly three level-m supertiles, any two of which share a side ending there (Theorem 2.1(b)); (iii) so the actual junctions are exactly the children's vertices at which the neighbour in Table 2 changes, each side being wholly shared with one sibling or wholly on the rim; (iv) A_c is placed by Lemma 6.4, and $B_c = A_c + \omega_m$ by hypothesis; (v) the rim of the parent is given by Table 3 (Proposition 2.4(c)). Hence the correspondence holds, and with it the marks on each child's rim, their order, the runs, partners, lengths and the order of the rim runs are those of $\mathcal{S}_T(\ell_m, d_m, \omega_m)$.

Lemma 7.1 (Parity Lemma)**.** *Say that (λ, d) satisfies the* order condition *if $\lambda(x) \geq 3$ for every letter x and $3 \leq d \leq \lambda(\delta) - 3$. If (λ, d) and (λ', d') both satisfy it, and $\lambda \equiv \lambda'$, $d \equiv d' \pmod 2$, then $\mathcal{S}_T(\lambda, d, \omega) = \mathcal{S}_T(\lambda', d', \omega)$ for every T and ω.*

Proof. We use two facts: a vertex of one child that lies on the rim of another is a vertex of that one too (in the abstract composition sides are glued end to end, so this holds by construction), and every composition has exactly one Σ child (Table 1). We list the possible marks on a side s of a child, of length ℓ, by their distance from its first vertex. A junction is a vertex, at distance 0 or ℓ. A crossover corner at vertex 0 of its child is also at distance 0 or ℓ, and the corresponding B lies at distance 1 or $\ell - 1$, on side 0 or 5 of its child or on the side glued to it. The crossover corners of the Σ child lie at distances d and $d \pm 1$ along its side 5, which has letter δ, or at distances $\ell - d$ and $\ell - d \mp 1$ along the side glued to it. A point inside a side lies only on that side and the side glued to it. So which of these *symbolic positions* carry marks is decided by Tables 1 and 2, by ω, and by which child is the Σ, and not by λ or d. Under the order condition, the symbolic positions that can occur on one side satisfy the strict inequalities

$$0 < 1 < d-1 < d < d+1 < \ell-1 < \ell \qquad \text{and} \qquad 0 < 1 < \ell-d-1 < \ell-d < \ell-d+1 < \ell-1 < \ell,$$

where in both cases $\ell = \lambda(\delta)$ if d occurs. Hence the cyclic order of the marks on each child's rim, and which marks coincide, are the same for (λ, d) and (λ', d'); matching marks with the same symbolic positions is a bijection as in the definition of equal skeletons (Section 5). The partner runs and the order of the rim runs are then determined by Tables 2 and 3. Finally, the length of a run is a sum of side lengths plus a difference of two symbolic positions, so its parity is determined by λ mod 2 and d mod 2. □

7.2. **The induction.**

Computation 7.2 (Base case and step)**.**
(a) Every level-1 supertile has the signature at (A, B), where A is as in Lemma 6.4 and $B = A + \omega_1$ in own order; and ℓ_1, d_1 are as stated in Proposition 6.1 and Lemma 6.4.
(b) For every kind T and $\omega \in \{\pm 1\}$, the reduced walk of Proposition 5.4 on $\mathcal{S}_T(\ell_1, d_1, \omega)$ closes no loop and gives hooks at all parent marks except the Γ child's A and B, which carry the crossover $\mathrm{R}(A) \leftrightarrow \mathrm{R}(B)$, $\mathrm{L}(A) \leftrightarrow \mathrm{L}(B)$.

Part (a) was computed from the tiles of the built level-1 supertiles (see after Lemma 2.6), using Lemma 4.2, in exact arithmetic and again in floating point. For (b), the 18 skeletons were built from Tables 1 to 3 and the lengths alone, without geometry, by two separate programs (one reads the tables as printed, the other the pattern extracted from the built supertiles, which is checked equal to them; Appendix A). Each has 8 children (7 for Γ), 24 marks (21) and 52 runs (45), of which 10 (9) lie on the rim.

Theorem 7.3 (Signature Theorem)**.** *In every homochiral tiling by* Tile$(1, 1)$*, every level-m supertile of every kind has the signature at (A, B), where A is its crossover corner (Section 2.3; placed as in Lemma 6.4 for $m \geq 1$) and $B = A + \omega_m$ in its own order. In particular no supertile contains a closed strand.*

Proof. Induction on m. For $m = 0$ this is Lemma 4.4, and for $m = 1$ it is Computation 7.2(a). Let $m \geq 1$, assume the theorem at level m, and let Y be a level-$(m + 1)$ supertile of kind T. Its children have the signature at $(A_c, A_c + \omega_m)$, so the skeleton of its composition is

$\mathcal{S}_T(\ell_m, d_m, \omega_m)$, by (**). By Lemma 6.5 both (ℓ_m, d_m) and (ℓ_1, d_1) satisfy the order condition, and they are congruent mod 2, by Corollary 6.3 and since $d_m = 2F_{3m+1}$ is even (Lemma 6.4). So, by Lemma 7.1, the skeleton of Y is $\mathcal{S}_T(\ell_1, d_1, \omega_m)$, and by Proposition 5.4 and Computation 7.2(b), Y has no closed strand and has hooks at all its rim vertices except the crossover corners A_Γ, B_Γ of its Γ child, with the crossover $\mathrm{R}(A_\Gamma) \leftrightarrow \mathrm{R}(B_\Gamma)$, $\mathrm{L}(A_\Gamma) \leftrightarrow \mathrm{L}(B_\Gamma)$ there. These two corners are adjacent on the rim of Y, since sides 0 and 5 of the Γ child lie on the rim (Table 2). By definition A_Γ is the crossover corner of Y, placed as in Lemma 6.4. Finally $B_\Gamma = A_\Gamma + \omega_m$ in the child's own order, which is opposite to that of Y; so $B = A - \omega_m = A + \omega_{m+1}$ in the own order of Y. □

Remark 7.4. Since the own orientation alternates with m and so does ω_m, in absolute terms B is always the counter-clockwise neighbour of A on the rim, as it is for the tile ($B = v_1$).

7.3. **No closed strands.**

Definition 7.5 (Faults)**.** The hierarchy of a homochiral tiling *covers the plane* if the supertiles containing some (equivalently, every) tile exhaust the plane. The union of the supertiles containing a given tile is an *infinite supertile*; if the hierarchy does not cover the plane, there are at least two, and the union of their boundaries is the *fault*.

(The two forms are equivalent, since two tiles that lie in a common supertile have the same supertiles from that level on; for the same reason two infinite supertiles are equal or have disjoint interiors.) Tilings with faults exist (Section 9). None of this is used in the proofs below. A closed strand is finite, but in a tiling with a fault it need not lie in a single supertile, so Theorem 7.3 alone does not exclude it. The argument below treats all tilings alike: a closed strand that is short compared with the sides of the level-m supertiles, and crosses between them, would have to turn back somewhere; only a crossover can make it turn back; and the crossovers of different supertiles are far apart. Recall that $\beta_m = F_{3m+1} + 2$ is the shortest side length at level m (Lemma 6.5).

Lemma 7.6 (Localisation Lemma)**.** *Let C be a closed strand of a homochiral tiling, with r pieces, and let $m \geq 1$ with $\beta_m > r$. Suppose that C does not lie in a single level-m supertile. Then the rim edges across which C passes from one level-m supertile to another form a tree G with at most r edges, which lies in one side or in the union of the three sides at one supertile vertex. So the tiles of C lie in two level-m supertiles that share a side, or in three around a common vertex.*

Proof. Follow C and cut it at the tile edges where it passes from one level-m supertile to another. There is at least one, so C is a cyclic sequence of arcs of level-m supertiles Y (Lemma 4.1), joined across *crossing edges*, each of which lies on the rims of two different supertiles. Each arc is a hook or a crossover arc of Y (Theorem 7.3), so the two crossing edges at its ends are rim edges of Y with a common vertex: for a hook at u they are the two rim edges at u, and for a crossover arc they are two of the three rim edges at A and $B = A \pm 1$. Hence the crossing edges form a connected graph G with at most r edges. It lies in the *rim graph*, the union of the rims of all level-m supertiles, which by Theorem 2.1(b) is a graph in which the vertices of the supertiles have degree 3, all other rim vertices degree 2, and any path between two different supertile vertices runs along a whole side, of at least $\beta_m > r$ edges. So G contains at most one supertile vertex u, and lies in one side, or in the union of the three sides at u. Every cycle of the rim graph contains a supertile vertex, since the other rim vertices lie inside sides, which are paths; leaving that vertex along a side, the cycle runs to the far end of the side. So every cycle has more than r edges, and G is a tree. Every level-m supertile containing a tile of C is entered by C across a crossing edge on its rim. If G lies in one side, the only such supertiles are the two that share it. Otherwise they are among the three supertiles around u, each side at u being shared by two of them. □

Lemma 7.7 (Lonely Crossover Lemma)**.** *Let $m \geq 2$, and let $Y \neq Y'$ be level-m supertiles of a homochiral tiling, with crossover corners A_Y and $A_{Y'}$ (Section 2.3). Every path in the rim graph from A_Y to $A_{Y'}$ has at least β_m edges.*

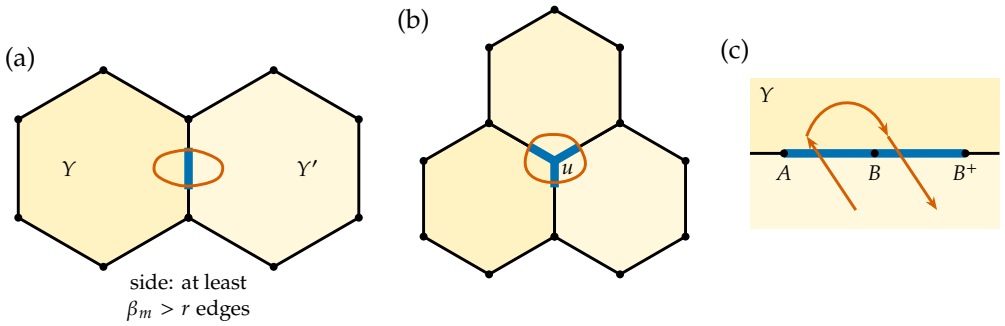


FIGURE 10. The Localisation Lemma, with supertiles drawn schematically as hexagons. A closed strand with r pieces (vermilion) passes between supertiles only across a connected set of at most r rim edges (blue), while every side has more than r edges. So the crossings lie (a) inside one side, between the two supertiles that share it, or (b) near one vertex u, among the three supertiles around it. (c) A ribbon (vermilion) crosses the edge AB into Y towards A, the crossover arc of Y carries it back towards B, and it leaves across the next edge towards B^+: the walk of Lemma 7.8 runs B, A, B, B^+ and turns back at A (in the notation of the lemma, $X = A$, $Z = B$, $Z' = B^+$).

Proof. Write $A = A_Y$ and $A' = A_{Y'}$. As in the proof of Lemma 7.6, the supertile vertices are the vertices of degree 3 of the rim graph, and a point inside a side lies on exactly two level-m supertiles, the two that share the side. Adjacent supertiles share whole sides with matching labels, that is, the same letter with opposite signs, or both η (Theorem 2.1(b), [17, Fig. 4.2]). By Table 4, side 0 of every kind other than Σ is labelled α^- or γ^+ and side 5 is labelled β^- or ζ^-, and side 5 of Σ is labelled δ^+; no two of these labels match.

Suppose first that neither Y nor Y' is a Σ. By Lemma 6.4, A and A' are vertex 0 of Y and of Y'. If $A = A'$, then Y and Y' are two of the three supertiles at this vertex, so they share a side ending there (Theorem 2.1(b)); it is side 0 or 5 of each, and the labels do not match. So $A \neq A'$, and a path from A to A', leaving the supertile vertex A along a side, runs along that whole side before it can reach another supertile vertex; it has at least β_m edges.

Suppose now that Y is a Σ; the case of Y' is symmetric. By Lemma 6.4, A lies inside side 5 of Y, at distances d_m and $\delta_m - d_m$ from its ends, and both are at least β_m (Lemma 6.5; here $m \geq 2$ is used). A path from A runs along side 5 of Y until it reaches A' or an end of that side. In the second case it has at least β_m edges. In the first, A' is an interior point of side 5 of Y, so it is not a supertile vertex: Y' is a Σ with A' inside its side 5, and Y' is the supertile across side 5 of Y. Then side 5 of Y is glued to side 5 of Y', and both are labelled δ^+, which do not match. □

Lemma 7.8 (Hairpin Lemma)**.** *In the situation of Lemma 7.6, follow C and record a walk W in G: crossing an edge xy of G from the port near x to the port near y (the straight-through rule) is a step from x to y, and a crossover arc of a supertile Y, which joins a port near A_Y to a port near B_Y (Definition 4.3), is a step between A_Y and B_Y along the edge A_YB_Y. Then W is a closed walk of positive length in G, and:*

(a) W turns back, *that is, leaves a vertex along the edge by which it arrived, only at a crossover corner A_Y or B_Y of a supertile Y whose crossover arc C traverses;*
(b) *if C traverses a crossover arc of Y, then A_YB_Y is an edge of G, and W visits a vertex whose distance in the tree G from the midpoint of A_YB_Y is $\frac{3}{2}$.*

Proof. C crosses at least one edge, so W has positive length; its crossing steps run along edges of G, and so do its crossover steps, as shown below. Between two crossings C runs along one arc of a level-m supertile Y, a hook or a crossover arc (Theorem 7.3). A hook at a rim vertex u joins the ports near u on the two rim edges at u; so W arrives at u along one of

these edges and leaves along the other, and does not turn back. Let $A = A_Y$ and $B = B_Y$. In absolute terms each crossover arc of Y joins the port of the edge AB near one of its ends, X say, to the port near the other end, Z, on the other rim edge ZZ' at Z: for $B = A + 1$ in the own order of Y, the arc R(A)–R(B) has $X = A$ and $Z = B$, and the arc L(A)–L(B) has $X = B$ and $Z = A$; for $B = A - 1$ the two arcs exchange these roles. When C traverses such an arc from X to Z, it arrives at the port near X on AB by crossing AB from the port near Z, and it leaves from the port near Z on ZZ' by crossing ZZ'; so W runs through Z, X, Z, Z', and in the other direction through Z', Z, X, Z. In both cases W turns back at X and not at Z, the edges AB and ZZ' lie in G, and W visits Z', whose distance in G from the midpoint of AB is $\frac{3}{2}$ (the path from Z' to the midpoint runs through Z, since G is a tree). These are all the places where consecutive steps of W meet, so (a) and (b) follow. □

Proof of the Endless Strand Theorem. Let C be a closed strand of a homochiral tiling, with r pieces, and choose $m \geq 2$ with $\beta_m > r$. If C lies in one level-m supertile, Theorem 7.3 excludes it. Otherwise let G and W be as in the Localisation Lemma and the Hairpin Lemma (Lemmas 7.6 and 7.8). For any point o of the tree G, at a vertex of W farthest from o both neighbours of that vertex along W are the neighbour on the path towards o, so W turns back there. Hence, by Lemma 7.8(a), C traverses a crossover arc of some supertile Y. If it also traversed one of a supertile $Y' \neq Y$, then A_Y and $A_{Y'}$ would lie in G (Lemma 7.8(b)), joined by a path in G of at most $r < \beta_m$ edges, contrary to the Lonely Crossover Lemma (Lemma 7.7). So every crossover arc that C traverses belongs to Y, and W turns back only at A_Y or B_Y. Take for o the midpoint of A_YB_Y: the vertex of W farthest from o is A_Y or B_Y, at distance $\frac{1}{2}$ from o. But by Lemma 7.8(b), W visits a vertex at distance $\frac{3}{2}$ from o, a contradiction. So there is no closed strand, and by Lemma 3.4 every strand is a two-way infinite path. □

8. The nest tiling and the single strand

8.1. **The nest.** Child 1 of every supertile is a Δ (Table 1), and child 1 of a Δ is again a Δ.

Construction 8.1 (The nest)**.** Let $\mathcal{T}$ be a homochiral tiling [17, Thm. 2.2]. For each $h \geq 0$, let $Y^{(h)}$ be a level-h Δ-supertile of $\mathcal{T}$; one exists, for instance child 1 of any level-$(h+1)$ supertile. Let $Y^{(h)} = Y^{(h)}_h \supset Y^{(h)}_{h-1} \supset \cdots \supset Y^{(h)}_0$ be its chain of descendants, each child 1 of the next. Let g_h be the proper motion taking the tile $Y^{(h)}_0$ onto Tile(1, 1) (with v_0 at the origin), and put $N^{(h)}_b = g_h(Y^{(h)}_b)$ for $b \leq h$.

Proposition 8.2. *$N^{(h)}_b$ does not depend on $h \geq b$ (or on the choices made). Writing N_b for it, $N_0 = \{\text{Tile}(1,1)\}$, and N_b is child 1 of N_{b+1} for every b.*

Proof. Let $h, h' \geq b$. By Lemma 2.6 there is a proper motion f with $f(Y^{(h)}_b) = Y^{(h')}_b$ that maps descendants to descendants with their numbers; hence $f(Y^{(h)}_{b'}) = Y^{(h')}_{b'}$ for all $b' \leq b$, in particular for the tiles Y_0. So $g_{h'}f$ and g_h both take the tile $Y^{(h)}_0$ onto Tile(1, 1), and since a tile has no symmetry they are equal. Thus $N^{(h')}_b = g_{h'}f(Y^{(h)}_b) = N^{(h)}_b$. □

We call N_b the *nest block* of level b, or simply a *block*. The *nest tiling* is $\mathcal{N} = \bigcup_{b\geq 0} N_b$, a set of tiles. Figure 11 shows three consecutive blocks.

Lemma 8.3 (Grandparent Lemma)**.** *For every $b \geq 1$, the block N_b does not meet the rim of N_{b+2}.*

Proof. By Table 2, the sides 0, 1, 3, 4, 5 of child 1 face the siblings 7, 0, 2, 3, 4, and side 2 lies on the parent's rim. A point of N_b that is not on its closed side 2 is therefore an interior point of N_{b+1}. This is clear for interior points of N_b and for interior points of the other sides, where N_b and one sibling together fill a neighbourhood. At the vertices 0, 1, 4, 5 of N_b, exactly

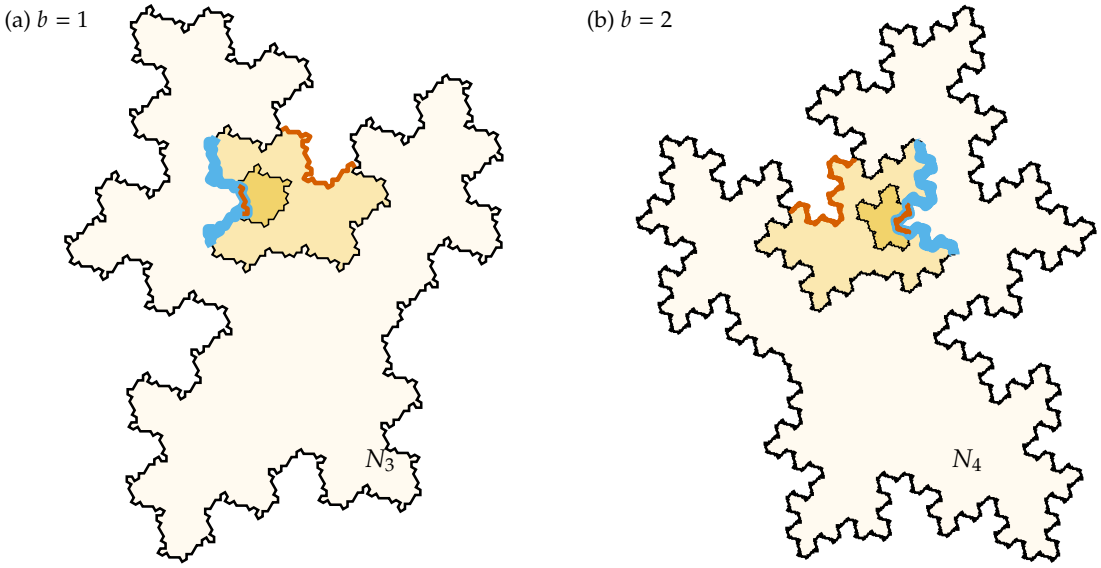


FIGURE 11. Three consecutive blocks of the nest: (a) $N_1 \subset N_2 \subset N_3$ and (b) $N_2 \subset N_3 \subset N_4$. Vermilion: side 2 of the two smaller blocks, which is the only side of each block on the rim of the next. Blue: side 4 of the middle block. Since the vermilion side of the smallest block lies inside the blue side, and the blue side faces a sibling inside the largest block, the smallest block does not reach the rim of the largest (Lemma 8.3).

three level-b supertiles meet; two of them are the distinct siblings across the adjacent sides, and together they fill a neighbourhood. Hence

$$N_b \cap \partial N_{b+1} = \text{closed side 2 of } N_b. \tag{2}$$

By Table 3 (the Δ row: positions 14 to 18), side 4 of the parent is the concatenation of $2.2, 2.1, 1.2, 0.4, 0.3$. Since 1.2 is neither first nor last, the closed side 2 of N_b lies in the relative interior of side 4 of N_{b+1}. Now let $p \in N_b \cap \partial N_{b+2}$. Then $p \in \partial N_{b+1}$, for otherwise p would be interior to $N_{b+1} \subset N_{b+2}$. So p lies on side 4 of N_{b+1} by the above, and, by (2) for N_{b+1}, on the closed side 2 of N_{b+1}. But the closed sides 2 and 4 of N_{b+1} are disjoint: they are paths on a simple closed curve, separated by side 3 in one direction and by sides $5, 0, 1$ in the other. □

Proposition 8.4. *$\mathcal{N}$ is a homochiral tiling of the plane, and any finitely many tiles of $\mathcal{N}$ lie in a common block N_b.*

Proof. The blocks are nested, so any finitely many tiles of $\mathcal{N}$ lie in a common block; in particular any two have disjoint interiors. We show that the tiles cover the plane. Fix h and work in the homochiral tiling $\mathcal{T}_h = g_h(\mathcal{T})$, which contains $N_0, \dots, N_h$. Call two tiles of $\mathcal{T}_h$ *neighbours* if they share a point, and let dist be the resulting graph distance. We claim that every tile at distance at most k from N_1 lies in N_{1+2k}, provided $1 + 2k \le h$. For $k = 0$ there is nothing to prove. If t has distance $k + 1$, it shares a point p with a tile of N_{1+2k}. By the Grandparent Lemma (Lemma 8.3), p is an interior point of N_{3+2k}, so the interior of t meets the interior of a tile of N_{3+2k}; hence t is that tile. Now let $x \in \mathbb{R}^2$ and let D be the disc of radius $r = |x| + 1$ about the origin. The tiles of $\mathcal{T}_h$ meeting D have diameter $3 + \sqrt{3} < 5$ and area $3 + 3\sqrt{3} > 8$, so there are at most $k_0 = \lfloor \pi(r + 5)^2/8 \rfloor$ of them. They cover the connected set D, so their neighbour graph is connected. One of them is $N_0 \subset N_1$, since the origin is a vertex of N_0. So every tile of $\mathcal{T}_h$ meeting D, in particular one containing x, lies in N_{1+2k_0} as soon as $h \ge 1 + 2k_0$. Since N_{1+2k_0} does not depend on h, x is covered by $\mathcal{N}$. □

b	tiles in N_b	arcs of N_b	$k=1$	$k=2$	$k=3$	$k=4$
0	1	14	5	2	1	1
1	9	46	10	2	2	1
2	71	182	37	2	1	1
3	559	758	146	2	2	1
4	4,401	3,198	613	2	1	1
5	34,649	13,534	2,586	2	2	1
6	272,791	57,318	10,949	2	1	–
even $b \geq 2$				2	1	1
odd $b \geq 1$				2	2	1

TABLE 5. The number of arcs of N_{b+k} on which the arcs of N_b lie, that is, the number of parts of $\mathcal{P}_k(b)$ (row $b = 0$: the 14 pieces of the single tile, computed on the drawing up to level 5). Rows $1 \leq b \leq 6$: computed by following every ribbon, up to level 9 (in floating point; for $b \leq 3$ also in exact arithmetic, up to level 7). Last two rows: Theorem 8.5 and its proof, for every b.

The blocks are supertiles: for $h \geq b$, $N_b = g_h(Y_b^{(h)})$ is a level-b Δ-supertile of the homochiral tiling $\mathcal{T}_h = g_h(\mathcal{T})$, and it is child 1 of N_{b+1} there. Since the arcs and closed strands of a patch depend only on its tiles, Proposition 2.4 and Theorem 7.3 apply to the blocks. In particular $\mathcal{N}$ has no closed strand: a closed strand passes through finitely many tiles, which lie in a common block N_b by Proposition 8.4, and would be a closed strand of N_b.

8.2. **Merging.** For $b \geq 1$ and $k \geq 0$, every arc of N_b is contained in a unique arc of N_{b+k}, because N_{b+k} has no closed strand. Let $\mathcal{P}_k(b)$ be the partition of the $E_b(\Delta)$ arcs of N_b in which two arcs are in the same part when they lie on the same arc of N_{b+k}. The parts only merge as k increases.

Theorem 8.5 (Merging)**.** *For every $b \geq 1$, $\mathcal{P}_4(b)$ has a single part; if b is even, so does $\mathcal{P}_3(b)$.*

Table 5 shows the numbers. The proof follows the arcs of N_b through the levels using Proposition 5.4 and Corollary 5.5, and shows that this computation is the same for all b of the same parity.

Label states. Fix $b \geq 1$. We follow the arcs of N_b up the nest. By Theorem 7.3 every arc of N_{b+k} is a hook at a rim vertex other than the crossover corners, or one of the two crossover arcs.

Definition 8.6 (Label state)**.** Let $k \geq 0$ and $N = N_{b+k}$, with its rim ports taken in its own orientation. A *label state* at relative level k is a tuple (Λ, H, S, X), where

- Λ is a finite set of *labels*, each label ν standing for a non-empty set $a(\nu)$ of arcs of N_b, the sets of distinct labels being disjoint; for $L \subseteq \Lambda$ we write $a(L) = \bigcup_{\nu \in L} a(\nu)$;
- H is a finite set of pairs (u, L), with u a rim vertex of N other than A and B, the u distinct, and L a non-empty set of labels;
- S is a set of pairwise disjoint *stretches*: sequences of consecutive rim vertices of N containing neither A, nor B, nor any u with $(u, L) \in H$;
- $X = (X_{\mathrm{R}}, X_{\mathrm{L}})$, each entry a non-empty set of labels or empty;

and the label sets in H and X are pairwise disjoint, with union Λ. The state *represents* $\mathcal{P}_k(b)$ if (i) for every vertex of a stretch, the hook of N there contains exactly one arc of N_b; (ii) for $(u, L) \in H$, the arcs of N_b in the hook of N at u are exactly those in $a(L)$; (iii) the arcs of N_b in the crossover arc of N through $\mathrm{R}(A)$ (through $\mathrm{L}(A)$) are exactly those in $a(X_{\mathrm{R}})$ (in $a(X_{\mathrm{L}})$), and there are none if the entry is empty; (iv) every arc of N_b lies in exactly one of the sets of (i)–(iii).

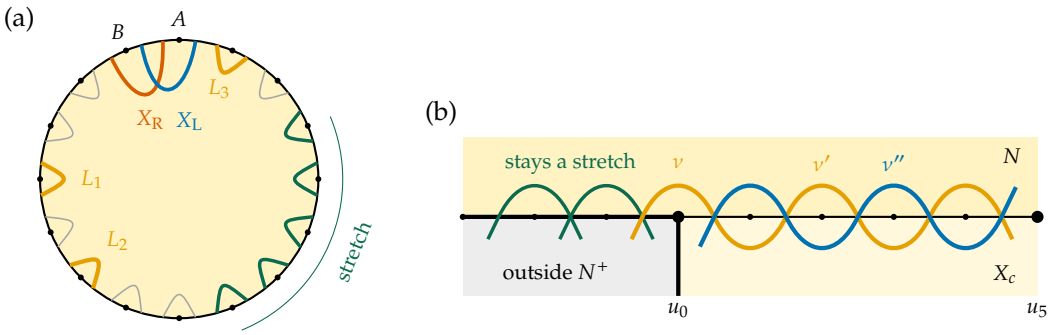


FIGURE 12. Label states. (a) A state on the rim of $N = N_{b+k}$, oriented counterclockwise: the green hooks are at the vertices of a *stretch*, and each contains exactly one arc of N_b; each orange hook carries a set L_i of labels and contains the arcs of N_b in $a(L_i)$; the crossover arcs carry X_R and X_L; the grey hooks contain no arc of N_b. (b) One step, with N as child 1 of N^+. A stretch on N's rim is split at the mark u_0 (S2). The part inside a rim run stays a stretch of N^+. The hook at the mark gets a fresh label v. Along the interior run from u_0 to u_5, shared with a sibling X_c, the hooks zigzag onto two arcs of N^+ (Corollary 5.5): those at even offsets get one fresh label v', those at odd offsets another, v''. Here v and v' lie on the same arc of N^+, so (S4) puts them in one label set.

If a state represents $\mathcal{P}_k(b)$ (Fig. 12), the parts of $\mathcal{P}_k(b)$ are the sets $a(L)$, one for each *labelled arc* of N (an arc recorded in H or X), and, for each stretch vertex, the singleton holding the one arc of N_b in its hook, so

$$|\mathcal{P}_k(b)| = (\text{number of labelled arcs}) + (\text{number of stretch vertices}). \tag{3}$$

At $k = 0$ the state with two labels x_R, x_L standing for the two crossover arcs, $H = \varnothing$, one stretch consisting of all rim vertices of N_b other than A and B, and $X = (\{x_R\}, \{x_L\})$, represents $\mathcal{P}_0(b)$.

The step. Let a state at level k be given, and pass to $N^+ = N_{b+k+1}$, in which $N = N_{b+k}$ is child 1; by (**) (whose hypothesis holds by Theorem 7.3), the composition is the abstract one for $(\ell_{b+k}, d_{b+k}, \omega_{b+k})$, with its positions, and its skeleton is $\mathcal{S}_\Delta(\ell_{b+k}, d_{b+k}, \omega_{b+k})$. Positions and ports on the rim of N are those of child 1 of N^+ in its own orientation, which is that of N, measured from its vertex 0. Fresh labels are added to Λ.

(S1) For $(u, L) \in H$: if u is a mark of child 1, put L at the reduced port $R_1(u)$, which lies on the hook of child 1 at u; if u lies inside a rim run of child 1, put (u, L) into H^+; if u lies inside an interior run, put L at the reduced port named in Corollary 5.5.

(S2) Split each stretch at the marks of child 1 that it contains. Each such mark u gets a fresh label, put at $R_1(u)$ and standing for the arc of N_b in the hook at u. Each maximal part between these marks lies inside one run. If it is a rim run, the part is put into S^+. If it is an interior run, then by Corollary 5.5 the hooks of the part lie on at most two arcs of N^+, one for each parity of their offsets in the run; for each parity that occurs, a fresh label, standing for the arcs of N_b in those hooks, is put at the reduced port named there. Which parities occur depends only on whether the part has one vertex or more, and on the parity of its first offset.

(S3) Put X_R at $R_1(A_1)$ and X_L at $L_1(A_1)$ (if not empty).

(S4) Every reduced port lies on an arc of N^+: applying σ and τ alternately from it, in either direction, reaches a parent port, since N^+ has no closed strand (Proposition 5.4 and Theorem 7.3). To each arc of N^+ assign the union of the label sets put at reduced ports on it. Put into H^+ the pair (u, L) for each parent mark $u \neq A^+, B^+$ whose hook in N^+ is assigned a non-empty set L; and let X_R^+ (X_L^+) be the set assigned to the crossover

arc of N^+ through $\mathrm{R}(A^+)$ (through $\mathrm{L}(A^+)$), the ports taken in the own orientation of N^+, or empty if that set is empty.

Lemma 8.7. *If the state at level k represents $\mathcal{P}_k(b)$, the state that the step produces represents $\mathcal{P}_{k+1}(b)$.*

Proof. By Proposition 5.4 and Corollary 5.5, each hook and crossover arc of child 1 lies on the arc of N^+ named in (S1)–(S4); the walk ends at a parent port, since N^+ has no closed strand (Theorem 7.3). A hook of child 1 at a vertex inside a rim run is a hook of N^+ at the same vertex (Proposition 5.4(a)), so carried pairs and carried stretch vertices keep properties (ii) and (i). A label put at a reduced port lies on an arc of N^+ that ends at parent ports (Proposition 5.4(b)); by Theorem 7.3 such an arc is a hook at a parent mark other than A^+, B^+, or a crossover arc, so every labelled arc of N^+ is recorded in H^+ or X^+ (hooks at the other rim vertices hold no reduced port; they are the carried pairs and stretches); a parent mark is not inside a rim run of child 1, so the pairs added in (S4) are at vertices different from the carried ones, and from the carried stretches. Every element of the old state is placed exactly once, so (iv) persists, and the new label sets, the unions over the arcs of N^+, are disjoint with union Λ^+. □

Lemma 8.8 (Uniformity in b)**.** *For $b \geq 1$ and $1 \leq k \leq 4$, the label state at relative level k, with every position written as an integer affine function of the letter lengths $x = \ell_b \in \mathbb{Z}^8$ at level b, depends on b only through the parity of b.*

Proof. Positions. By Proposition 6.1, every side length at level $b + k$ is a letter length, and (1) gives ℓ_{b+k} as an integer linear function of x; and $d_{b+k} = \zeta_{b+k}$, since both are $2F_{3(b+k)+1}$ (Proposition 6.2 and Lemma 6.4). So every position used in the steps is an integer affine function of x: the rim positions of the vertices and marks of the children in their own order, the offsets of labelled hooks and stretch endpoints inside runs, and the positions in N^+ of carried points. The skeletons depend only on the parity of b, by Lemmas 6.5 and 7.1, and parities of positions are fixed by Corollary 6.3 and Lemma 6.4.

Decisions. Besides these, the steps make the following decisions that depend on the lengths: (1) on which side of a child a position lies; (2) whether a crossover corner of a child is a vertex of that child; (3) the bounds $d_{b+k} \geq 2$ and $\ell_{b+k}(\Sigma, 5) - d_{b+k} \geq 2$; (4) whether two candidate mark positions coincide; (5) the strict cyclic order of the marks on each child's rim, and of the marks of child 1 inside a stretch; (6) whether a point is a mark of child 1, or lies strictly inside a given run, and at which offset; (7) whether a mark lies in a stretch; (8) whether a part of a stretch is empty, and whether it is a single vertex; (9) whether a parent mark is the parent's corner A^+; (10) whether the positions of labelled hooks are distinct; (11) the reduction of a position modulo a rim length. Each is an inequality $f(x) \geq t$ or an equality $f(x) = 0$ for an integer affine function f; a disequality is recorded as $\pm f(x) \geq 1$, and a reduction of a position p modulo a rim length E, with a fixed quotient q, as $0 \leq p - qE$ and $p - qE < E$. The program also makes comparisons that it does not record, because their outcome is fixed by recorded ones: whether $B = A + \omega$ must be reduced modulo the rim, which for children other than Σ depends only on the sign of ω and for Σ never happens, by the bounds (3); two tests in the map from a child's rim positions to the parent's, decided for the non-negative positions passed to it by the decisions (1); and the matching of candidate mark positions with the marks of the skeleton, and the search for the run that contains a point, whose outcomes are fixed by the skeleton and the decisions (2) and (4)–(6). At each decision the procedure for a given b evaluates a predicate that is determined by the outcomes of the earlier decisions. So if every recorded inequality and equality holds for b, then, by induction on the number of decisions, the procedure for b takes the same branches as the symbolic run below and produces the same label states.

The symbolic run. We ran the computation symbolically, carrying every position as an integer affine function of x, once for each parity of b. At each decision the branch was chosen by

evaluating at the actual letter lengths of $b = 2$ for even b and of $b = 1$ for odd b. Over 4 relative levels this records 583 distinct obligations for even b and 564 for odd b (Appendix A).

Certificates. Every recorded obligation holds for every $b \geq 1$, by Lemma 6.5: at level b every letter length is at least 5, and $\delta_b - \zeta_b = \delta_b - d_b \geq 4$. Indeed, write an inequality as $a \cdot x + e \geq t$ and let $c = \max(0, -a_\zeta)$, where a_ζ is the coefficient of ζ. For every recorded inequality, $a' = a - c\,(\mathbf{e}_\delta - \mathbf{e}_\zeta)$ has no negative coefficient and $5\sum a' + 4c + e \geq t$, so $a \cdot x + e = a' \cdot x + c\,(\delta_b - \zeta_b) + e \geq t$; for all but 7 inequalities of each parity, $c = 0$. Every recorded equality holds identically. So for every $b \geq 1$ every recorded obligation holds, and by the first two paragraphs the label states are those of the symbolic run of the same parity. (A second implementation of the whole run is described in Appendix A; the proof does not rely on it.) □

Proof of Theorem 8.5. By Lemma 8.7 and induction on k the label states represent $\mathcal{P}_k(b)$, and by Lemma 8.8 they are those of the symbolic run of the same parity. In those runs the number of labelled arcs at relative levels $1, 2, 3, 4$ is $4, 2, 1, 1$ for even b and $3, 2, 2, 1$ for odd b, and there is no stretch from relative level 2 on. By (3), $\mathcal{P}_3(b)$ (even b) and $\mathcal{P}_4(b)$ (odd b) have a single part. (At $k = 1$ a stretch is still carried, and its vertices give singleton parts; this is why the numbers of Table 5 are larger there, for instance $10 = 3 + 7$ for $b = 1$ and $37 = 4 + 33$ for $b = 2$.) The direct computations for $b \leq 6$ give the same numbers. Once $\mathcal{P}_k(b)$ has a single part, so does $\mathcal{P}_{k'}(b)$ for every $k' \geq k$. □

8.3. **One strand.** We now have everything needed for the second main theorem.

Proof of the Ariadne Theorem. By Lemma 3.4, every component of the strand graph $\mathcal{G}(\mathcal{N})$ is a cycle or a two-way infinite path, and there are no cycles, as shown after Proposition 8.4. It remains to show that $\mathcal{G}(\mathcal{N})$ is connected. Let p, q be pieces of tiles t, t' of $\mathcal{N}$. By Proposition 8.4 both tiles lie in some N_b with $b \geq 1$. Since N_b has no closed strand, p and q lie on arcs of N_b. By Theorem 8.5 these arcs lie on one arc of N_{b+4}, so p and q are on the same strand. The strand passes through every tile, since every tile has pieces. □

9. Scope and open questions

Other tilings. The Endless Strand Theorem holds for every homochiral tiling. The Ariadne Theorem is proved for the nest tiling, whose chain of supertiles is $\Delta \subset \Delta \subset \cdots$ with child number 1 at every step. Every tiling whose hierarchy covers the plane has an analogous chain $(K_m, c_m)_{m \geq 1}$: the level-m supertile X_m containing a fixed tile has kind K_m and is child c_m of X_{m+1}. The induction of Section 7 does not see the chain, but the merging of Section 8 does: its finite part depends on the sequence (K_m, c_m), and so may its outcome.

Remark 9.1 (Three strands: a computation, not a theorem). Let $K_m = \Theta$, $c_m = 2$ for odd m and $K_m = \Gamma$, $c_m = 7$ for even m (Table 1). A tiling with this chain is constructed as in Construction 8.1, and its hierarchy covers the plane (the argument of Lemma 8.3 and Proposition 8.4 applies, with Tables 2 and 3). Following the ribbons in exact arithmetic, the arcs of X_1 lie on exactly three arcs of X_m for $3 \leq m \leq 8$, and so do those of X_2 and X_3 for $5 \leq m \leq 8$; the label bookkeeping of Section 8 along this chain agrees, and still gives three at $m = 20$. If this persists, the pieces of this tiling form exactly three strands, each two-way infinite by the Endless Strand Theorem. We have not proved this, and the paper does not use it. For the other alignment (Γ at odd levels, Θ at even ones) the arcs of X_1 lie on a single arc of X_4. Along 90,000 chains chosen at random step by step (30,000 for each $b \leq 3$: the kind of X_b, and then at each level the parent's kind and the child number, uniformly among those allowed by Table 1), the arcs of X_b came to lie on a single arc of X_m for some $m \leq 20$ in all but five. The label bookkeeping along a chain (`hierarchy/chains_label.js`) agrees with exact strand following on 60 such chains up to level 8. The programs are in `experiments/`, with logs.

(1) Which homochiral tilings whose hierarchy covers the plane carry a single strand? Is the number of strands always finite, and which numbers occur?
(2) In a tiling with a fault, how many strands are there? The Endless Strand Theorem rules out closed strands there, but says nothing about their number.
(3) Which other decorations of Tile(1, 1) have no closed strand in any tiling, or a single strand in the nest tiling? The method applies to any decoration in which the tile has the signature at some edge (see below).

Faults. Some homochiral tilings contain two or more infinite supertiles, by the usual compactness argument: a limit of copies of a tiling moved so that the origin is a vertex of a level-m supertile, for $m = 1, 2, \ldots$, has the origin on the boundary of a supertile of every level (its supertiles are the limits of those of the copies, by the uniqueness in Theorem 2.1(a)). This is not used in the proofs. The Endless Strand Theorem covers these tilings: no strand closes up, across a fault or anywhere else. The Ariadne Theorem says nothing about them.

Other drawings and designs. The theorems concern only the pairing of ports (Definition 3.1) and the straight-through rule. They therefore apply to any drawing that realises these, including drawings on the curved Spectre, whose tilings correspond to the homochiral tilings by Tile(1, 1) [17, §2]. The method uses the decoration only through the signatures of the tile and of the Mystic (Lemma 4.4), the position of the crossover, which fixes A (Lemma 6.4), the base case and step (Computation 7.2), Lemma 7.7 for the Endless Strand Theorem, and the computations of Section 8 for the Ariadne Theorem. For another decoration in which the tile has the signature at some edge, these computations can be run again, but some arguments need re-checking, not just re-running: Lemma 6.4, the Parity Lemma (which uses that only the Σ child has its crossover corner inside a side) and Lemma 7.7 depend on where the crossover corners lie. The outcome may differ.

Appendix A. The computations

All programs and their logs are published with this paper at `https://cortado-j.github.io/SPECTRE_KNOT/`, folder `code/`. One command, `./run_all.sh`, re-runs everything the proof needs in about twelve minutes on a four-core machine, except the two comparisons with figures of [17], which need its PDF (not included) and run when the variable `SMKGS_PDF` names it; `./run_all.sh --all` adds the corroborating computations to level 9, and takes about an hour in all (with the memory of the JavaScript programs capped at 6.5 GB). The file `code/README.md` maps every statement of this paper to the script that checks it. (The folder `referee/` holds checks written during the AI system's review passes.)

Two implementations. The skeleton walks, the symbolic run and the certificates work with integers and rationals only; the integers involved are below $2 \cdot 10^7$ in absolute value, far below 2^{53}, so the parts written in JavaScript are exact as well. The geometric computations that the proof uses are the Mystic (Lemma 4.4) and the level-1 data and signatures (Proposition 6.1 and Lemma 6.4, Computation 7.2(a)). They are done in exact arithmetic by `exact/exact.py`, in Python. It represents every vertex exactly as an element $a + b\rho + c\rho^2 + d\rho^3$ of $\mathbb{Z}[\rho]$, $\rho = e^{i\pi/6}$, with integers a, b, c, d, and every motion as $z \mapsto \rho^k z + t$ (or $\rho^k \bar{z} + t$ in reflected frames). Floating point is used only to read off the sign of the area of a polygon whose area exceeds 8 in absolute value. The 18 skeleton walks of Computation 7.2(b) are done by `hierarchy/induction.js` and, separately, by `faults/faults.py`, in Python; the symbolic run of Lemma 8.8 by `hierarchy/certnest.js` and, separately, by `second/nest2.py`, in Python, which takes the pattern from the tables of this paper and shares no code with the first (see below). The two JavaScript programs that the proof uses (`induction.js`, `certnest.js`) read the pattern, ℓ_1 and the matrix M from files written by the JavaScript geometric computations below; `exact.py` checks these equal to the ones it extracts in exact arithmetic, and those are compared with Tables 1 to 4 as printed (`second/check_exact_tables.py`, through the transcription in `second/nest2.py`, which `second/check_tex.py` compares with the source

of this paper); the side letters that `certnest.js` reads are compared with Table 4 in the same way. All the programs were written by the same AI system.

The JavaScript programs in `core/` and most of `hierarchy/` repeat the geometric computations independently and extend them: the patterns for $m \le 7$, the signature up to level 9, the merging for $b \le 6$ up to level 9. Apart from the inputs just described, which are checked, these computations corroborate the proof but are not used in it. The programs share no code with `exact.py`, although both implement the key-point construction of [17, App. A] and so the same table of placements. They work in double-precision floating point and identify points by rounding coordinates to 10^{-2} or 10^{-3}. We do not bound their rounding error. A separate check (`referee/overlap.js`), which builds the supertiles of levels 1 to 7 in the same way, finds distinct vertices at distance at least 0.89, and points that it identifies within 4×10^{-12} of each other; and up to level 9 every rim found has exactly E_m edges (Proposition 6.2) and every rim vertex has degree 2, which a wrongly identified point would break. All programs compose the ribbon pairings with Lemma 4.2 and nothing else. (In the programs the sign ω_m is called `beta`.)

Further checks of the signature. The signature was also computed directly from the tiles of built supertiles, in exact arithmetic at levels 1 to 4 and in floating point at levels 1 to 9, for all nine kinds; at levels 1, 2, 3 the rims have 44, 160 and 652 edges for Γ, and 46, 182 and 758 for the other kinds. The skeletons obtained from (ℓ_2, d_2), (ℓ_3, d_3) and $(\ell_1 + 2\ell_2, d_1 + 2d_2)$ are identical to those from (ℓ_1, d_1) with the same ω, as Lemma 7.1 predicts, and the 18 skeletons of Computation 7.2(b) coincide with those extracted from the built supertiles at every level from 2 to 9 (this comparison does not distinguish the parent's A from its B). As a check of Section 7.3, `faults/faults.py` computes, symbolically for every level $m \ge 1$ with certified obligations, every pair of supertiles glued along a side (264 ordered pairs) and every triple around a vertex (489 ordered stars) allowed by the matching rule. None of these unions contains a closed strand, except three Γ-supertiles around a point that is vertex 5 of each (m odd) or vertex 1 of each (m even). By the argument of Section 7.3, a closed strand there passes the corners A of at least two of the three (Lemma 7.8(b)), each at distance β_m (m odd) or α_m (m even) from that point, so its crossing edges number at least $2\beta_m$, too many for Lemma 7.6 at level m. Unions glued by letter but against the signs of the matching rule do contain closed strands. The patterns of the compositions $m \to m + 1$ for $m \le 3$ were extracted in exact arithmetic and agree with Tables 1 to 3; for $m = 4$ the kinds and gluing of the children agree with Tables 1 and 2.

Checks of the composition pattern. The comparison with [17, Fig. 4.1] in the proof of Proposition 2.4 works on a render of the figure. For each cluster the script finds the six dots, fits our six vertices to them by a proper similarity with residual below 1% of an edge (any other matching, or the vertices of any other kind, fits at least 40 times worse), finds our whole rim on the printed outline, every inner edge of our tiles on a printed line and every printed line inside the cluster on one of our edges, and our first Mystic tile on the lighter shaded tile, and compares the labels, read from the figure by hand, with Table 4.

For the first check of Remark 2.5, at levels $m \le 4$ we used the vertices observed as the points where three level-m supertiles meet inside larger supertiles. For $m = 5, \dots, 8$ we propagated them by the rule of Proposition 2.4(c), learnt at the observed level of the same parity; at levels 5 and 6, where vertices were also observed, the two agree (for Ψ, at the four of its six vertices that were observed). So for $m = 6, 7$ the rule (c) is an input of the extraction rather than something it tests. For the second check, the kinds, positions and edge labels of the nine panels of [17, Fig. 5.1] were encoded by hand, and the marked vertices (with the centre of each panel's Γ hexagon) by a script from a render of the figure, which re-runs from the PDF. The script also confirms that the side-to-label dictionary (*) is the same in all 71 placements of a child. Two deliberately corrupted encodings (two outer labels exchanged; one wrong label in [17, Fig. 4.2]) were rejected. We also checked that the tiles of every

supertile of levels 1 to 7 (up to 2,147,679 tiles) have disjoint interiors and exactly fill the region bounded by its rim.

Further checks of the merging. In the proof of Lemma 8.8 the symbolic run records 583 distinct obligations for even b and 564 for odd b; for each parity one is an equality, which holds identically, and 7 use $\delta_b - \zeta_b \geq 4$.

The second implementation of the symbolic run, `second/nest2.py`, also takes the eight letter lengths at level b as variables and its branches at $b = 2$ and $b = 1$, but makes its comparisons differently: it records 142 and 144 distinct obligations and certifies them in the same way, by Lemma 6.5. It reaches the same numbers of labelled arcs, checks at every step that the labels and stretches account for every arc of N_b exactly once, finds that the concrete runs for $1 \leq b \leq 40$ take the branches of the symbolic run of their parity, and rejects deliberately corrupted inputs and false obligations.

At every relative level, the labels that the symbolic run places on the reduced ports of child 1, and the partition of the labels, agree with those of a run rebuilt from the composition pattern, the parities and ω alone; and that run agrees with the same data read off the built blocks N_b, $b \leq 6$, at every relative level up to level 9. The merging for $b = 1$ was also found by following every ribbon of the built blocks in exact arithmetic (`exact/exact.py`), which gives 10, 2, 2, 1, and by the label computation on label inputs read off the built blocks (`core/nest.js`), which gives 3, 2, 2, 1 labelled arcs with a stretch carried at $k = 1$ only, as does the label computation with the actual positions (`hierarchy/nest_b1.js`). The other direct computations of the merging are described in the caption of Table 5.

Sizes. The script `hierarchy/sizes.py` computes these numbers from Table 1 and Proposition 6.2. They agree with the supertiles built by the programs (tile counts up to level 7, rim lengths up to level 9).

level m	1	2	3	4	5	6	7	8	9
tiles in a Δ	9	71	559	4,401	34,649	272,791	2,147,679	16,908,641	133,121,449
rim $E_m(\Delta)$	46	182	758	3,198	13,534	57,318	242,790	1,028,462	4,356,622
shortest side β_m	5	15	57	235	989	4,183	17,713	75,027	317,813

What was computed.

statement	computation	script
Definition 3.1 and Lemma 4.4	the tile's pairing; the Mystic, exact and in floating point	`core/tile.js`; `exact/exact.py`, `hierarchy/checks.js`
Proposition 2.4, Remark 2.5	vertices of levels 1–6 (points where three supertiles meet); the pattern of compositions $m \to m+1$, $m = 1, \ldots, 7$, all identical; exact for $m \leq 3$	`hierarchy/vertices.js`, `combin.js`, `exact/exact.py`
Tables 1 to 4	the printed tables equal the patterns, side lengths and Σ offsets extracted in exact arithmetic (levels 1–4); children of the built supertiles of levels 2–5 glued as in Table 2	`second/check_exact_tables.py`, `second/check_tex.py`, `exact/exact.py`
Proposition 2.4 (base of $(*)$)	the level-1 supertiles as hat/turtle clusters, fitted to [17, Fig. 4.1] (dots, outline, inner tile edges, shading; labels read by hand)	`referee/fig41_compare.py`, `fig41_render.py`
Remark 2.5	agreement with an encoding of [17, Fig. 5.1], all nine kinds; its dots re-found from the PDF; two corrupted encodings rejected	`referee/fig51_compare.py`, `fig51_raster.py`
Remark 2.5	tiles of every supertile, levels 1–7, have disjoint interiors and fill the region bounded by its rim	`referee/overlap.js`
Propositions 6.1 and 6.2, Corollary 6.3, and Lemma 6.4	letters, recurrence, lengths at levels 1–8, closed forms	`hierarchy/recur.js`, `letters.py`

statement	computation	script
Comp. 7.2(a)	signature at level 1 (and directly at levels 2–4): exact, and in floating point	`exact/exact.py`, `hierarchy/checks.js`
Comp. 7.2(b)	the 18 skeleton walks, by two programs; parity lemma corroborated; equal to the skeletons of the built supertiles at levels 2–9 (the computed skeletons record slightly more than Definition 5.3, e.g. rim-run parities)	`hierarchy/induction.js`, `faults/faults.py`
Proposition 5.4 (check)	reduced walk equals direct composition at all 1,424 parent ports, levels 1–8	`hierarchy/checks.js`
Lemma 8.8	symbolic label computation in the letter lengths at level b, 583 + 564 distinct obligations, all certified by Lemma 6.5	`hierarchy/certnest.js`
Lemma 8.8 and Theorem 8.5	second implementation: symbolic label computation, 142 + 144 distinct obligations, certified by Lemma 6.5; concrete runs for $1 \le b \le 40$ follow the same branches; Table 5 reproduced for $1 \le b \le 6$; corrupted inputs rejected	`second/nest2.py`
Theorem 8.5 ($b = 1$, corroboration)	the label computation for $N_1, \dots, N_5$ with the actual positions: $10, 2, 2, 1$ parts	`hierarchy/nest_b1.js`
Theorem 8.5, Table 5	strand following in exact arithmetic for $b \le 3$ to level 7, in floating point for $b \le 6$ to level 9; label computation on inputs from the built blocks for $b \le 6$ to level 9, and the same inputs rebuilt from the pattern, the parities and ω (corroboration); row $b = 0$ on the drawing	`exact/exact.py`, `core/hier2.js`, `core/nest.js`, `hierarchy/synthnest.js`, `drawing/onetile.js`
corroboration	signature and skeleton periodicity at levels 1–9, all kinds; 0 closed strands at every level	`core/reduce.js`, `hierarchy/synth.js`
Section 7.3 (corroboration)	every pair of supertiles glued along a side and every triple around a vertex allowed by the matching rule, symbolically for all levels, certified: no closed strand except in three Γ-supertiles around their vertex 5 or 1, whose strand is too long for Lemma 7.6 at that level; configurations against the matching rule (closed strands found)	`faults/faults.py`
Section 3 (drawing)	the drawn ribbons realise the pairing, levels 1–3; shared edges glue e_i to e_j with $i + j$ odd, levels 1–4	`drawing/checkmodel.js`, `exact/exact.py`
Remark 3.6	the drawn pieces: containment, the single crossing, angles at the edges	`drawing/drawcheck.py`

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